

PI3K α inhibition blocks osteochondroprogenitor specification and the hyper-inflammatory response to prevent heterotopic ossification

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eLife Assessment

This study, which includes additional experiments in response to the reviewer comments, presents **valuable** findings illustrating the role of PI3K α in heterotopic ossification in FOP model mice. The methods, data, and analyses are **solid** and generally support the claims although as noted by one of the reviewers, there is no data demonstrating the effect of BYL79 on cell growth, and it remains unclear whether BYL79 also inhibits the Smad2/3 pathway. While this study provides new insights into the role of the PI3K α pathway as a therapeutic target for FOP, questions about the mechanism of BYL79 still exist.

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Abstract

Heterotopic ossification (HO) occurs following mechanical trauma and burns, or congenitally in patients suffering from fibrodysplasia ossificans progressiva (FOP). Recently, we demonstrated that inhibitors of phosphatidylinositol 3-kinase α (PI3K α) may be a useful therapy for patients undergoing HO. In this study, using the already marketed BYL719/Alpelisib/Piqray drug, we have further confirmed these results, detailed the underlying mechanisms of action, and optimized the timing of the administration of BYL719. We found that BYL719 effectively prevents HO even when administered up to three to seven days after injury. We demonstrate in cell cultures and in a mouse model of HO that the major actions of BYL719 are on-target effects through the inhibition of PI3K α , without directly affecting ACVR1 or FOP-inducing ACVR1^{R206H} kinase activities. *In vivo*, we found that a lack of PI3K α in progenitors at injury sites is sufficient to prevent HO. Moreover, time course assays in HO lesions demonstrate that BYL719 not only blocks osteochondroprogenitor specification, but also reduces the inflammatory response. BYL719 inhibits the migration, proliferation and expression of pro-inflammatory cytokines in monocytes and mast cells,

suggesting that BYL719 hampers the hyper-inflammatory status of HO lesions. Altogether, these results highlight the potential of PI3K α inhibition as a safe and effective therapeutic strategy for HO.

Introduction

Heterotopic ossification (HO) is a disorder characterized by ectopic bone formation at extraskeletal sites, including skeletal muscle and connective tissues. Trauma-induced HO develops as a common post-operative complication after orthopedic surgeries (e.g., hip arthroplasty), blast injuries, deep burns and nervous system injuries (Hwang et al., 2022 [↗](#)). In addition to trauma-induced HO, fibrodysplasia ossificans progressiva (FOP) is a rare congenital autosomal dominant disorder also involving HO (Bravenboer et al., 2015 [↗](#)). Ectopic bones form progressively through endochondral ossification, mostly in episodic flare-ups associated with inflammation (Towler and Shore, 2022 [↗](#)). These cumulative osteogenic events lead to reduced mobility resulting from ankylosing joints. Affected individuals have a shorter life-span, most commonly due to thoracic insufficiency syndrome (Kaplan et al., 2010 [↗](#); Pignolo et al., 2016 [↗](#)).

HO reflects a shift from a normal tissue repair process to an aberrant reactivation of bone-forming developmental programs that require acute inflammation and the excessive expansion of local progenitors, followed by inappropriate differentiation of these progenitors into chondroblasts and osteoblasts which finally results in bone formation (Hwang et al., 2022 [↗](#)). In both trauma-induced HO and FOP, pathology appears following the excessive activation of receptors sensitive to Transforming growth factor- β (TGF β) superfamily members (Lees-Shepard et al., 2018 [↗](#); Wang et al., 2018 [↗](#)). FOP arises from gain-of-function mutations in the bone morphogenetic protein (BMP) type I receptor *ACVR1*, with the most common mutation being c.617G>A, R206H (Shore et al., 2006 [↗](#)). Higher SMAD1/5 mediated signaling of mutated *ACVR1* has been partially attributed to a loss of auto-inhibition of the receptor and mild hypersensitivity to BMP ligands (Chaikuad et al., 2012 [↗](#); Groppe et al., 2011 [↗](#); Van Dinther et al., 2010 [↗](#)). Signaling by mutated *ACVR1* still relies on ligand-induced heterotetrameric clustering but, unlike wild type *ACVR1*, does not require *ACVR2A/B* kinase activity (Agnew et al., 2021 [↗](#); Ramachandran et al., 2021 [↗](#)). More importantly, mutated *ACVR1* receptors alter their signaling specificity, abnormally activating SMAD1/5 and altering the non-canonical BMP signals (e.g., p38 or PI3K) in response to activin A (Hatsell et al., 2015 [↗](#); Hino et al., 2015 [↗](#); Valer et al., 2019a [↗](#)). Accordingly, evidence supports that activin A is necessary and sufficient for HO in FOP (Hatsell et al., 2015 [↗](#); Lees-Shepard et al., 2018 [↗](#); Upadhyay et al., 2017 [↗](#)). In contrast, activin A does not drive post-traumatic HO (Hwang et al., 2020 [↗](#)), and non-genetically-driven HO mostly arises from excessive BMP and TGF β signaling, with functional redundancy between different type I receptors of the TGF β -superfamily (Agarwal et al., 2017 [↗](#); Patel et al., 2022 [↗](#); Sorkin et al., 2020 [↗](#); Wang et al., 2018 [↗](#)).

Evidence points to mesenchymal fibroadipogenic precursors (FAPs), which are widely distributed in muscle and other connective tissues, as the key cell-of-origin that aberrantly undergo chondrogenesis and further ectopic bone formation (Dey et al., 2016 [↗](#); Eisner et al., 2020 [↗](#); Lees-Shepard et al., 2018 [↗](#)). However, both FOP or non-genetic HO also require local tissue destruction and inflammation, which indicate that FAPs should be invariably primed for ectopic bone formation by this inflammatory microenvironment (Barruet et al., 2018 [↗](#); Convente et al., 2018 [↗](#); Hwang et al., 2022 [↗](#); Matsuo et al., 2019 [↗](#); Pignolo et al., 2013 [↗](#); Sorkin et al., 2020 [↗](#)).

Inflammation in HO is characterized by an initial acute response following injury with the activation of innate immunity and the influx of neutrophils and monocytes (Convente et al., 2018 [↗](#); Hwang et al., 2022 [↗](#)). Additionally, B and T cells of the adaptive system are also recruited (Chakkalakal et al., 2012 [↗](#); Convente et al., 2018 [↗](#)). However, mice lacking B or T lymphocytes exhibited no delay in the development of heterotopic ossification (HO) after injury, indicating that these cells may play a subsequent role in the dissemination of bone lesions (Kan et al., 2009 [↗](#)).

During intermediate and late inflammatory stages, the recruitment of monocytes, macrophages and mast cells occurs, which in turn exerts autocrine and paracrine effects on nearby FAPs (Chakkalakal et al., 2012 [↗](#); Convente et al., 2018 [↗](#); Sorkin et al., 2020 [↗](#); Tu et al., 2022 [↗](#)). Highlighting their relevance in HO, the depletion of mast cells and macrophages profoundly impairs genetic and non-genetic HO (Convente et al., 2018 [↗](#); Torossian et al., 2017 [↗](#); Tu et al., 2022 [↗](#)). BMP receptors are robustly expressed in monocytes and macrophages and the expression of *ACVR1^{R206H}* has been shown to extend inflammatory responses in patient-derived macrophages (Barruet et al., 2018 [↗](#); Matsuo et al., 2021 [↗](#), 2019 [↗](#)). Among the plethora of cytokines secreted by monocytes, macrophages and mast cells, both activin A and TGF- β stand out as extremely relevant for HO (Alessi Wolken et al., 2017 [↗](#); Hatsell et al., 2015 [↗](#); Lees-Shepard et al., 2018 [↗](#); Patel et al., 2022 [↗](#); Sorkin et al., 2020 [↗](#); Upadhyay et al., 2017 [↗](#)).

Genetic and pharmacological studies have indicated that osteochondroprogenitor specification and maturation depend on phosphatidylinositol 3-kinase- α (PI3K α) (Ford-Hutchinson et al., 2007 [↗](#); Fujita et al., 2004 [↗](#); Gámez et al., 2016 [↗](#); Ikegami et al., 2011 [↗](#)). We found that PI3K signaling was also linked to HO, since inhibitors of PI3K α (BYL719/Alpelisib/Piqray) prevented HO in mouse models without major side-effects (Valer et al., 2019a [↗](#)). Mechanistically, PI3K α inhibitors hamper canonical and non-canonical BMP signaling, decreasing total and phosphorylated SMAD1/5 levels and reducing transcriptional responsiveness to BMPs/Activin A in mesenchymal progenitors (Gámez et al., 2016 [↗](#); Valer et al., 2019b [↗](#)). In the current study, we demonstrate that the pharmacological and genetic inhibition of PI3K α in HO progenitors at injury sites reduces HO *in vivo*. Moreover, envisioning a future translation into the clinic, we have optimized the administration of BYL719 and found that the delayed administration of BYL719, up to seven days after inflammatory injury, still prevents HO in mice. In addition, we found that BYL719 blocks osteochondroprogenitor specification and effectively reduces the essential inflammatory response. Altogether, the data presented here show the potent therapeutic effect of PI3K α inhibition on HO in mouse models.

Results

Delayed initiation of treatment with PI3K α inhibitor effectively prevents heterotopic ossification

Heterotopic ossification takes place following a precise temporal pattern of progenitor activation and the recruitment of distinct cell types at the ossification centers which causes early divergence from the normal skeletal muscle repair program. We previously found that the pharmacological administration of BYL719 prevents HO in a mouse model of HO (Valer et al., 2019a [↗](#)). Given that BYL719 is marketed for the treatment of cancer and overgrowth syndrome, it is a promising therapeutic molecule in pathological HO. To further determine a potential therapeutic window for BYL719 and better understand the cellular targets of BYL719, we evaluated the effect of BYL719 when administered intermittently and several days after HO induction. For this, we used a conditional mouse model (*ACVR1^{Q207Dfl/fl}*) in which we combine the overexpression of Cre recombinase through adenoviral particles with the injection of cardiotoxin intramuscularly in the hindlimb (Fukuda et al., 2006 [↗](#)). In this model, we administered intermittently BYL719 (i.p. 25 mg/kg) or vehicle control starting one, three or seven days after HO induction. We also implemented a different administration regimen of BYL719 only for the initial three days following the injury (**Figure 1A** [↗](#)). None of the treatments led to significant changes in mouse body weight (**Figure 1-Figure Supplement 1** [↗](#)). HO volume was analyzed by micro computed tomography (μ CT) twenty-three days post-injury. The early administration of the treatment with BYL719 at day one or three after the onset of HO resulted in significantly lower HO; late administration seven days after the induction of HO was also partially effective (**Figure 1B** [↗](#) and

C). Discontinuation of the treatment after three days did not prevent HO at day 23. These results indicate that the inhibition of PI3K α by BYL719 after injury prevents HO and this protection is still effective if the treatment is started several days after injury, expanding its therapeutic window.

Deficiency of PI3K α at injury sites is sufficient to partially prevent heterotopic ossification

Since other small molecules designed to target PI3K α have reported partial off-target effects on kinases other than PI3K α (Furet et al., 2013; Jamieson et al., 2011), we aimed to confirm that the results obtained with BYL719 were due to the specific inhibition of PI3K α . Therefore, we developed a conditional mouse model (*ACVR1^{Q207D}^{fl/fl};p110 α ^{fl/fl}*) in which the Cre recombinase simultaneously drives the expression of *ACVR1^{Q207D}* and the deletion of the catalytic subunit of PI3K α (*p110 α*) upon intramuscular hindlimb combined injection of adenovirus-Cre and cardiotoxin to trigger HO. With this approach, Cre mediates the expression of *ACVR1^{Q207D}* and the deletion of PI3K α in the same cells, whereas cells recruited afterwards will remain wild type for both genes. As expected, μ CT analysis performed 23 days after injury showed extensive HO in mice expressing *ACVR1^{Q207D}*, which was prevented by intermittent treatment with BYL719 initiated one day after injury (Figure 2A, 2B and 2D). Genetic deletion of *p110 α* in mice expressing *ACVR1^{Q207D}* led to a reduction in HO, which was further enhanced by BYL719 (Figure 2B). Importantly, no significant changes in the weight of the mice were observed, confirming the absence of severe toxicity (Figure 2-Figure Supplement 1A). Histomorphometrically, *ACVR1^{Q207D}* mice wild type for PI3K α developed islands and/or bone spurs with a well-organized bone structure. However, in PI3K α deficient mice *ACVR1^{Q207D}* expression only led to minor ectopic calcifications that were already surrounded by fully regenerated muscle tissue on the 23rd day after injury (Figure 2D, Figure 2-Figure Supplement 1B, Figure 2-Figure Supplement 2). Accordingly, analysis of the ratio bone volume/tissue volume (BV/TV) and the correlation between bone volume and BV/TV of individual mice was clearly different between *p110 α* genotypes, irrespective of BYL719 treatment (Figure 2C, Figure 2-Figure Supplement 1C). These results demonstrate that the abrogation of PI3K α activity (either genetic or pharmacologically) in cells at injury sites is relevant for their inhibitory effects in HO. This might be attributed to the direct effects of BYL719 on progenitor cell expansion and chondroblast specification, as well as its possible ability to reduce the inflammatory response required for HO.

Genetic or pharmacological inhibition of PI3K α prevents osteochondroprogenitor specification and reduces the number of FAPs at injury sites

Next, we aimed to assess the effects of PI3K α inhibition in osteochondroprogenitor cells. For this, we isolated bone marrow-derived mesenchymal stem cells (BM-MSCs) from *p110 α ^{fl/fl}* mice and transduced them *ex-vivo* with retrovirus expressing *Acvr1* (wild type and *R206H*) with or without Cre recombinase. The gene expression levels of exogenous mRNA of *Acvr1* and *Acvr1^{R206H}* was similar (Figure 3A). As anticipated, transduction with Cre viruses significantly reduced the expression levels of *p110 α* to approximately 40%. This is consistent with an estimated transduction efficiency of nearly 60% (Figure 3B). The overexpression of *Acvr1^{R206H}* increased basal and activin-dependent expression of canonical (*Id1* and *Sp7*) and activin-dependent expression of non-canonical (*Ptgs2*) BMP target genes (Figure 3C), known markers and drivers of osteochondrogenic differentiation (Nakashima et al., 2002; Wang et al., 2013). Both pharmacological and genetic inhibition of PI3K α reduced canonical and non-canonical *ACVR1^{R206H}* transcriptional responses induced by activin A. Of note, stimulation with activin A increased its own transcription (*Inhba*) in *Acvr1^{R206H}*-transduced BM-MSCs, which was also inhibited by BYL719 (Figure 3C). Therefore, BYL719 may also prevent osteochondroprogenitor differentiation indirectly, via transcriptional inhibition of *INHBA*/Activin A expression. These

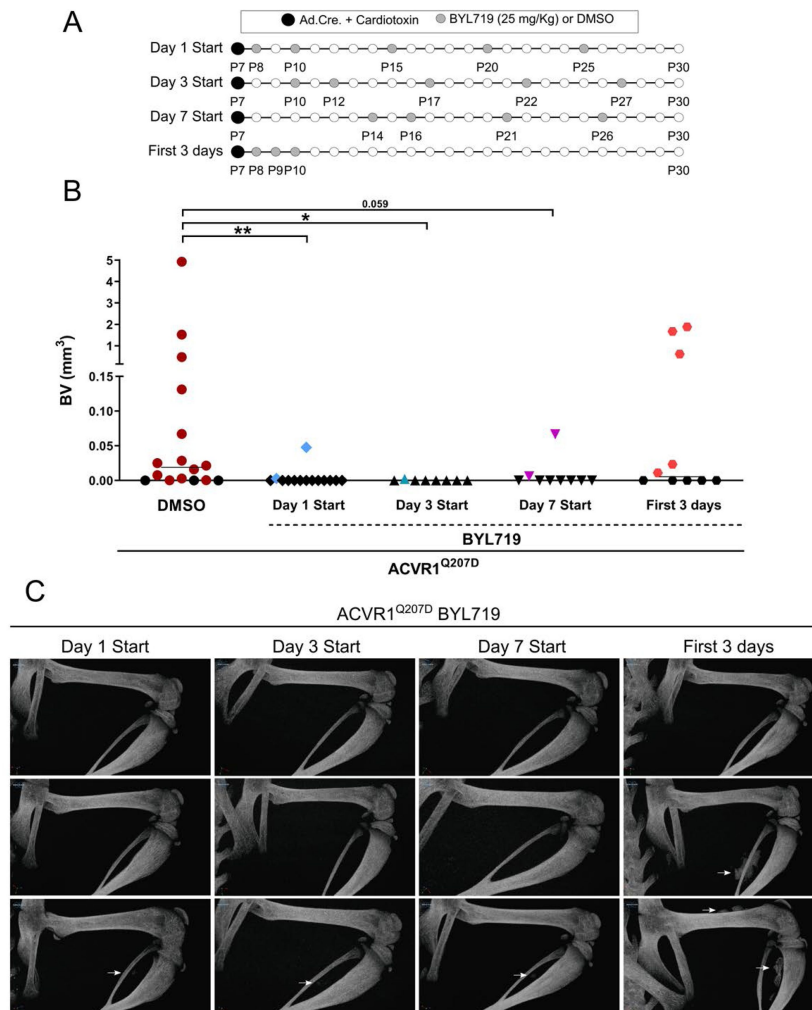


Figure 1.

(A) Heterotopic ossification was induced at P7 through the injection of adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. The pattern of 5 administrations of DMSO or BYL719 (25 mg/kg) is indicated by grey dots started at P8 (start day 1), P10 (delayed start day 3), and P14 (delayed start day 7). One group was administered DMSO or BYL719 (25 mg/kg) only at P8, P9, and P10 (first 3 days post-HO-induction), as indicated by grey dots. The final time-point, P30, was conserved between experimental groups. (B) Quantification of heterotopic ossifications bone volume (BV) (mm³) of each experimental group. Colored symbols indicate the presence of heterotopic ossifications. Black symbols indicate the absence of heterotopic ossifications. Individual mouse values with group median are shown. * $P < 0.05$, ** $P < 0.01$, Kruskal-Wallis test with Dunn's multiple comparison test. (C) Representative 3D microtomography images of the injected hindlimbs of 3 different mice for each experimental group. White arrows point to heterotopic ossification.

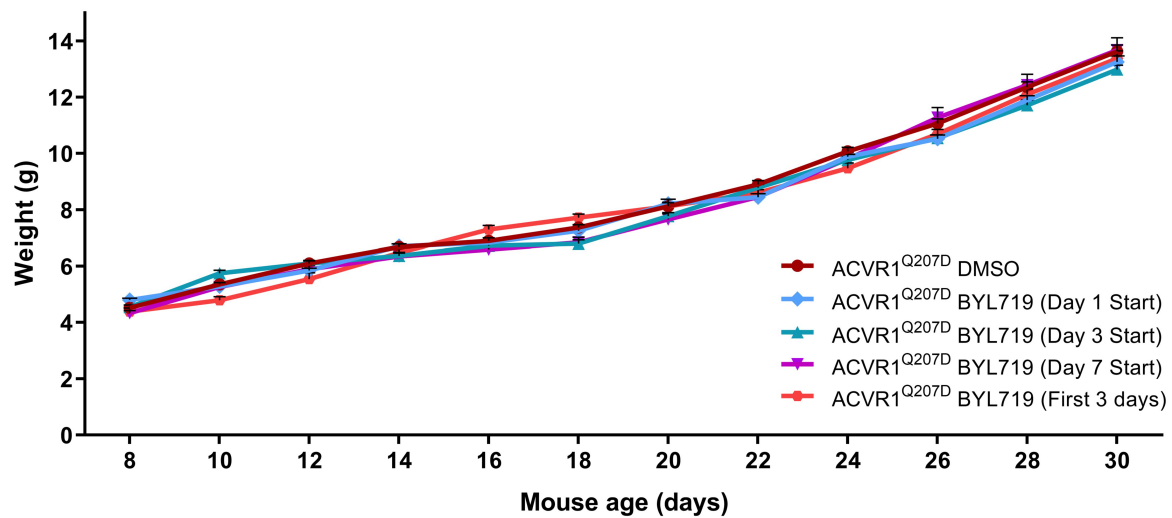


Figure 1-Figure supplement 1.

Mice body weight from P8 (one day post-HO-induction) to P30.

Data are shown as mean $\pm$ SEM. Two-way ANOVA with Sidak's multiple comparisons test.

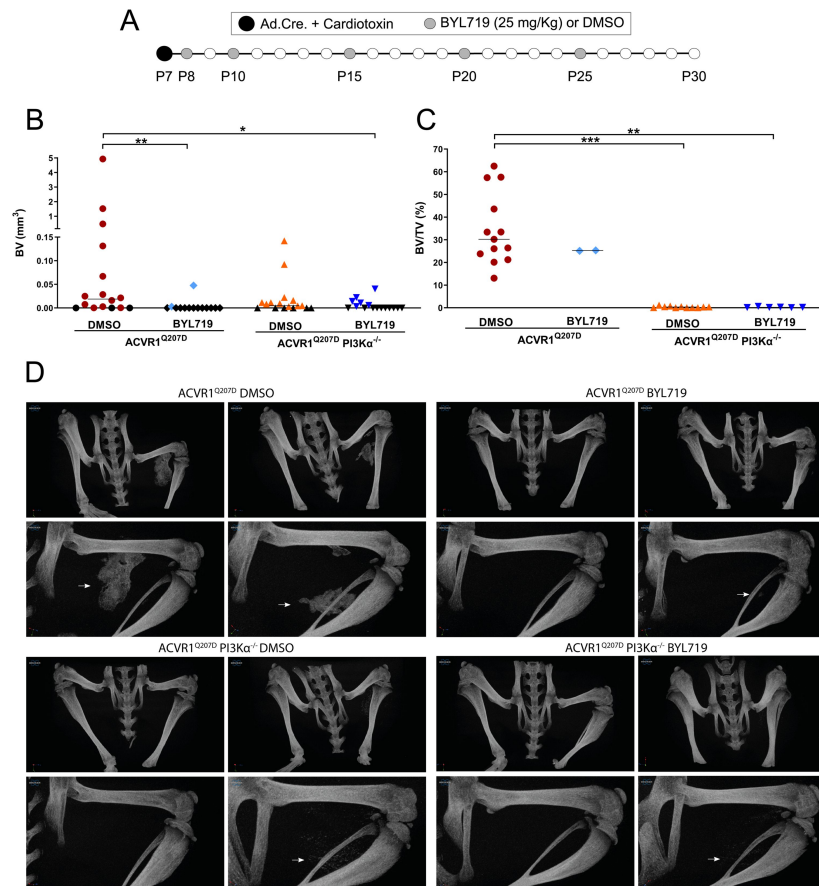


Figure 2.

(A) Heterotopic ossification was induced at P7 through the injection of Adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. Either DMSO (vehicle) or BYL719 (25 mg/kg) were injected following the scheme indicated with grey dots, starting at P8 (B) Quantification of heterotopic ossifications bone volume (BV)(mm³) of each experimental group. Colored symbols indicate the presence of heterotopic ossifications. Black symbols indicate the absence of heterotopic ossifications. Individual mouse values with group median are shown. *P<0.05, **P<0.01, Kruskal-Wallis test with Dunn's multiple comparison test. (C) Quantification of the ratio of bone volume per tissue volume (bone volume/tissue volume, BV/TV) within heterotopic ossifications of each experimental group, including only mice with detected HO. Data shown are of each individual mouse with the group median. **P<0.01, ***P<0.001, Kruskal-Wallis test with Dunn's multiple comparison test. (D) Representative 3D frontal microtomography images of the injected hindlimbs of mice for each experimental group and a detailed close-up image for each selected mouse. White arrows show heterotopic ossification in the close-up images.

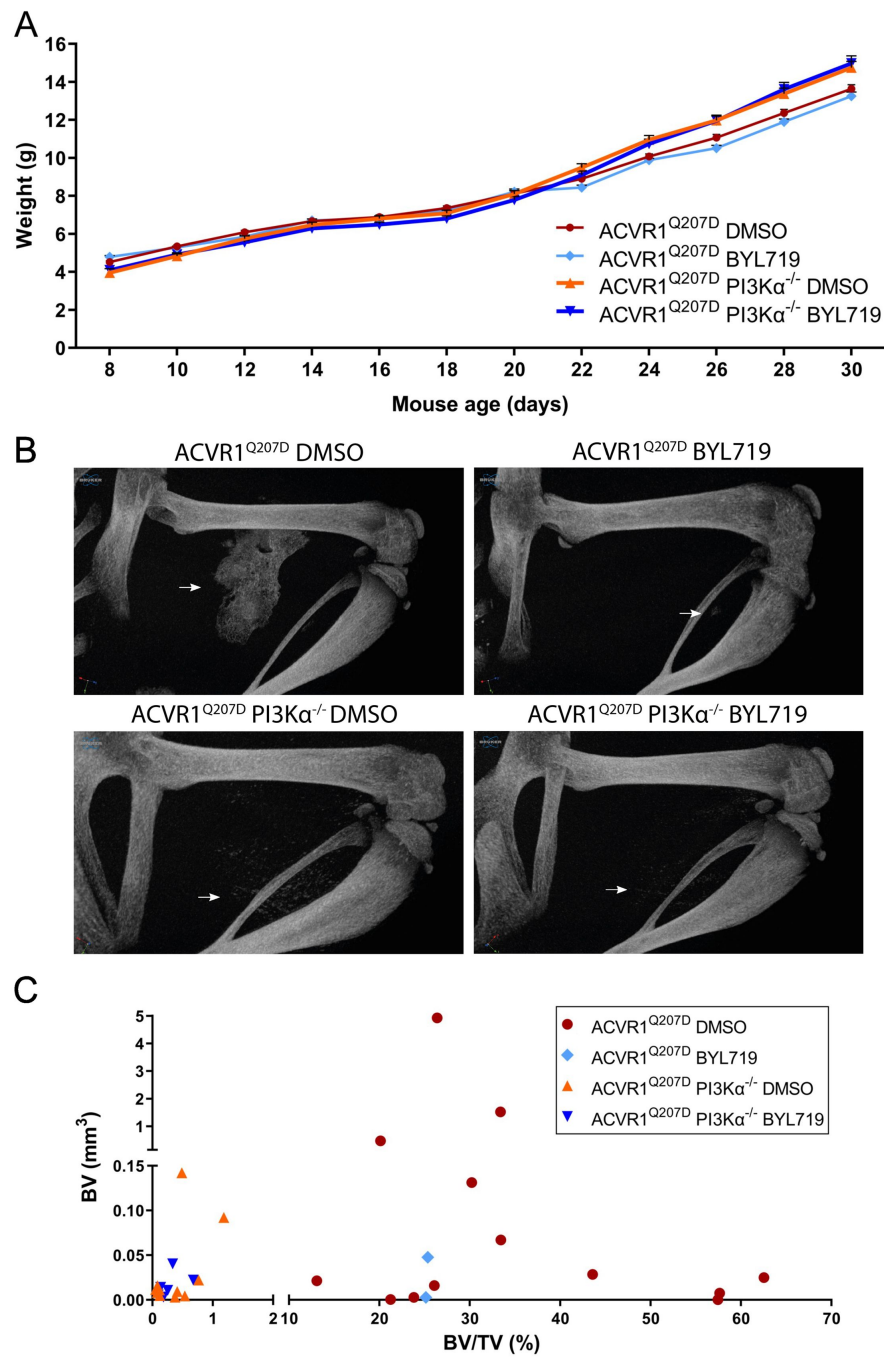


Figure 2-Figure supplement 1.

(A) Mice body weight from P8 (one day post-HO-induction) to P30 (final time-point). Data are shown as mean $\pm$ SEM. Two-way ANOVA with Sidak's multiple comparisons test. (B) Close-up 3D microtomography images of the injected hindlimbs of mice for each experimental group. White arrows show heterotopic ossification. (C) Two-dimensional representation of bone volume (BV) and bone volume/tissue volume (BV/TV) within heterotopic ossifications. Data shown are of each individual mouse with observed heterotopic ossification. Each symbol corresponds to an individual mouse. Colours indicate groups, as detailed in the legend.

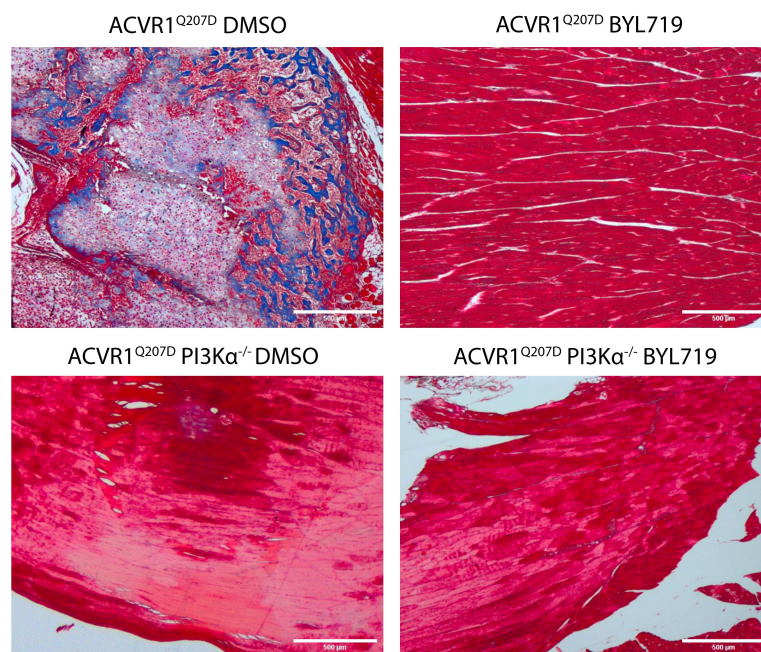


Figure 2-Figure supplement 2.

Muscle sections of mice 23 days after injury stained with Masson's Trichrome.

Different genotypes and treatments specified in the figure. All images were acquired at 4x magnification, (scale bar 500 µm).

results demonstrate a direct effect of BYL719 on osteochondroprogenitor cells and suggest that the effects of BYL719 on ACVR1-downstream signaling are mainly on target, due to PI3Kα kinase inhibition.

In addition, we analyzed the number of fibroadipogenic precursors (FAPs) during the progression of ectopic bone formation in our *in vivo* mouse model of HO. Samples of injured muscles of the conditional mouse model (*ACVR1^{Q207Dfl/fl}*) were subjected to immunofluorescence to identify the number of PDGFRA⁺ cells (FAPs) at 4, 9, 16 and 23 days after induction of HO. The number of FAPs remained sustained until day 16th after injury, decreasing afterwards (**Figure 3-Figure Supplement 1A, B**). More importantly, treatment with BYL719 reduced the number of PDGFRA⁺ cells throughout the ossification process. We also observed an increase in the diameter of myofibers in animals treated with BYL719 (**Figure 3-Figure Supplement 1A, C**). These results suggest an improved muscle regeneration in animals treated with BYL719 compared to those undergoing HO.

It is known that 2-aminothiazole-derived PI3Kα inhibitors could also inhibit ACVR1 kinase activity (Jamieson et al., 2011). Therefore, we investigated whether BYL719 effects could be explained by the reduction in ACVR1 kinase activity. For this, we made use of a kinase target engagement approach. Upon transient transfection of Nanoluciferase-tagged TGF-β receptors, we incubated the cells with BYL719 and/or an ATP-like tracer analogue. In the absence of an ATP competitor molecule, the tracer analogue and Nanoluciferase enzyme stay in close proximity, thereby allowing for bioluminescence resonance energy transfer (BRET) between the Nanoluciferase (energy donor, emitting at 460 nm) and the tracer analogue (energy acceptor). Once excited, the acceptor molecule emits fluorescence (610 nm) (**Figure 4A** and **B**). BYL719 incubated at 1 and 10 μM failed to significantly inhibit the acceptor emission of the BMP type I receptors ACVR1, BMPR1A, ACVR1B, TGFBR1 and the mutant receptor ACVR1^{R206H} (**Figure 4C**), without significant changes in the protein levels of each receptor (**Figure 4-Figure Supplement 1**). In addition, we demonstrated that BYL719 does not target the activity of the type II kinase receptors ACVR2A, ACVR2B, BMPR2 and TGFBR2. We confirmed these results using *in vitro* kinase activity assays with recombinant ACVR1^{R206H}. BYL719 exhibited no major effects on ACVR1^{R206H} activity up to 10 μM (**Figure 4D**).

Given that BYL719 does not directly target the activity of ACVR1 or related receptors, we established two *in vitro* models to further profile the main targeted pathways. In human MSCs (hMSCs) stably overexpressing ACVR1 or ACVR1^{R206H}, recombinant Activin A was able to induce SMAD1/5 activation and the expression of ACVR1 downstream targets *ID1* and *ID3* in cells overexpressing ACVR1^{R206H} (**Figure 5-Figure Supplement 1A and B**). We performed micromass chondrogenic differentiation assays in differentiation media supplemented with TGFβ1, with or without activin A and BYL719 in parental cells and cells transduced with ACVR1^{WT} or ACVR1^{R206H}. Coincubation with BYL719 (10 μM) completely inhibited the formation of a chondrogenic glycosaminoglycans-rich matrix in response to TGFβ1 plus activin A in the different cell types (**Figure 5A** and **B**) and partially abrogated the expression of the chondrogenic genes *ACAN* and *MMP13* (**Figure 5C**). Furthermore, in these cells, coincubation with BYL719 (10 μM) was able to repress Activin A-induced alkaline phosphatase activity upon 7 and 11 days of culture, respectively (**Figure 5D** and **E**). In addition, to confirm these results, we isolated MSCs from UBC-CRE-ERT2/*ACVR1^{R206H} fl/wt* knock-in mice. These cells, when treated with 4OH-tamoxifen, express the intracellular exons of human ACVR1^{R206H} in the murine *Acvr1* locus. Therefore, we can compare wild type and R206H MSCs isolated from the same mice. ACVR1^{R206H} mutant cells display an enhanced chondrogenic response to activin A compared to wild type cells (**Figure 5F**). In both wild type and mutant MSCs, treatment with BYL719 decreased the expression of chondrogenic. These results confirmed that BYL719 could inhibit chondrogenic differentiation of human and murine MSCs irrespective of the expression of either wild type or mutant ACVR1.

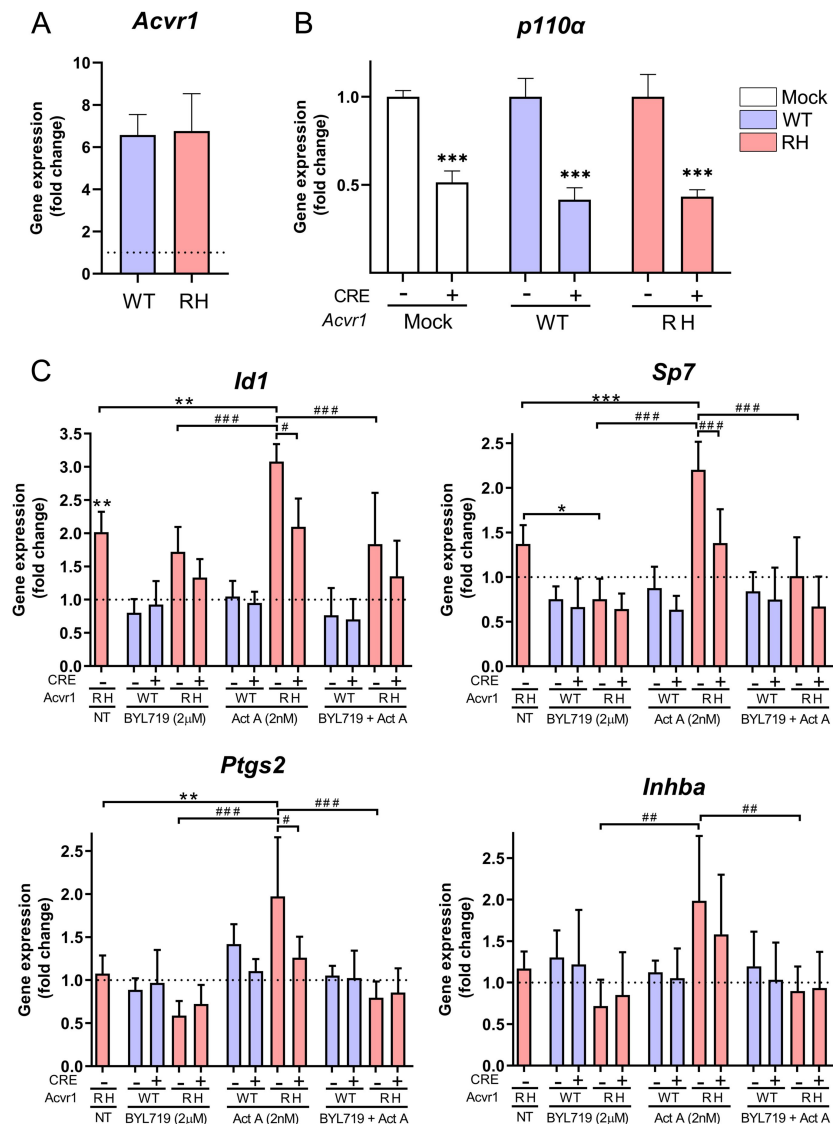


Figure 3.

Analysis of *Acvr1* gene expression by qPCR in BM-MSCs from *p110^{fl/fl}* mice infected with virus expressing wild-type *Acvr1* (WT) or *Acvr1^{R206H}* (RH).

(A) The endogenous expression level of *Acvr1* gene in mock-transfected BM-MSCs is shown as a dotted horizontal line. Data are shown as mean $\pm$ SD (n = 12 per group). Unpaired t-test between transfected groups. (B) Gene expression analysis of *p110α* in BM-MSCs from *p110^{fl/fl}* mice, infected with virus expressing wild-type *Acvr1* (WT) or *Acvr1^{R206H}* (RH) and/or Cre co-infection. Data are shown as mean $\pm$ SD (n = 3 per group). ***P<0.001, two-way ANOVA with Tukey's multiple comparisons test. (C) mRNA expression of canonical (*Id1* and *Sp7*), non-canonical (*Ptgs2*) target genes, and activin A (*Inhba*) in BM-MSCs *p110^{fl/fl}* transfected with *Acvr1* (wild type or R206H) with or without Cre recombinase. Cells were NT (not treated), or treated with BYL719 (2μM) and/or activin A (2nM). Expression data was normalized to those of control cells which were transfected only with *Acvr1* WT without any treatment, shown as a dotted horizontal line. Asterisks (*) refer to the differences between different conditions of *Acvr1* RH cells compared to control cells. Hash signs (#) refer to the differences between different conditions of *Acvr1* RH cells compared to *Acvr1* RH cells without Cre recombinase and treated with activin A. Data are shown as mean $\pm$ SD (n = 6 per group). * or # P<0.05, ** or ## P<0.01, *** or ### P<0.001, two-way ANOVA with Tukey's multiple comparisons test.

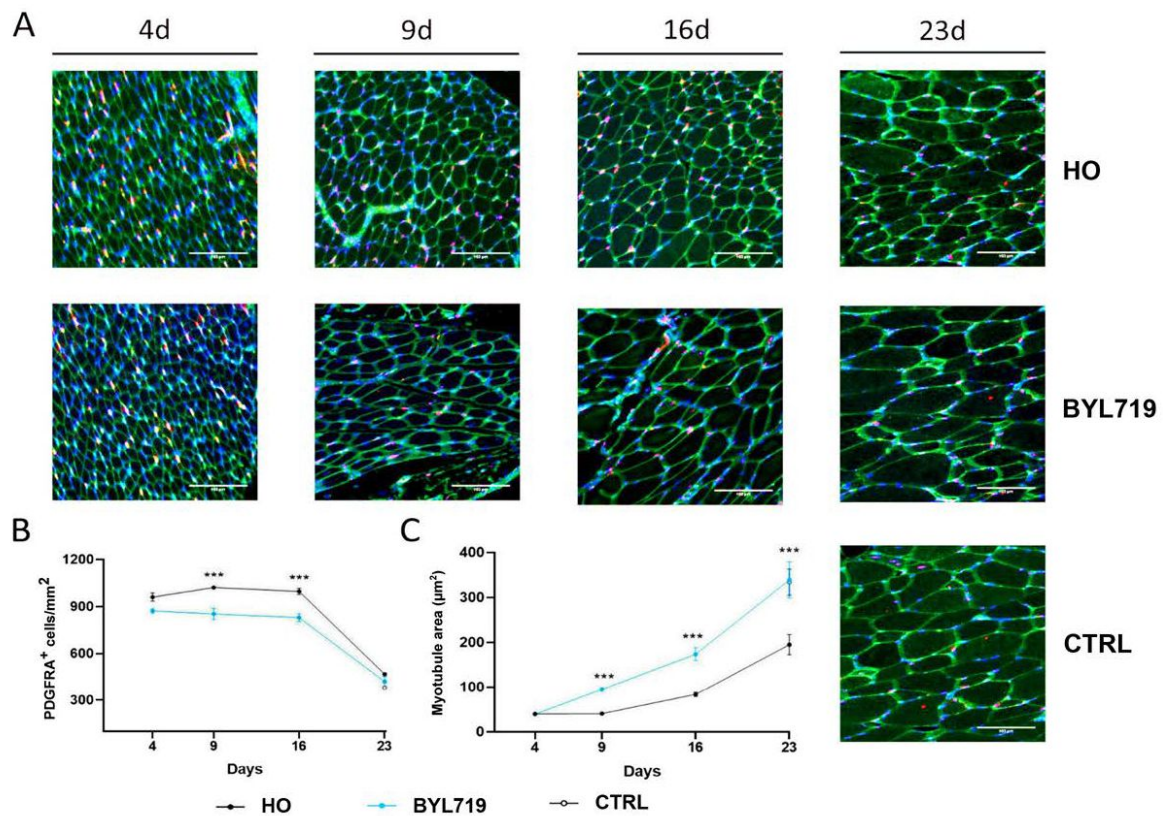


Figure 3-Figure supplement 1.

(A) Muscle sections of mice 4, 9, 16 and 23 days after induction of heterotopic ossification (HO). PDGFRA+ cells are indicated in red, wheat germ agglutinin staining is indicated in green and DAPI staining is indicated in blue. Different times and BYL719 treatment are specified in the figure. (B) Quantification of PDGFRA-positive FAPs obtained from five images obtained per mice, from 4 mice per group, for a total of 20 quantified images per group. (C) Quantification of myotube diameter estimated from wheat germ agglutinin staining obtained from five images per mice, from 4 mice per group, for a total of 20 quantified images per group.

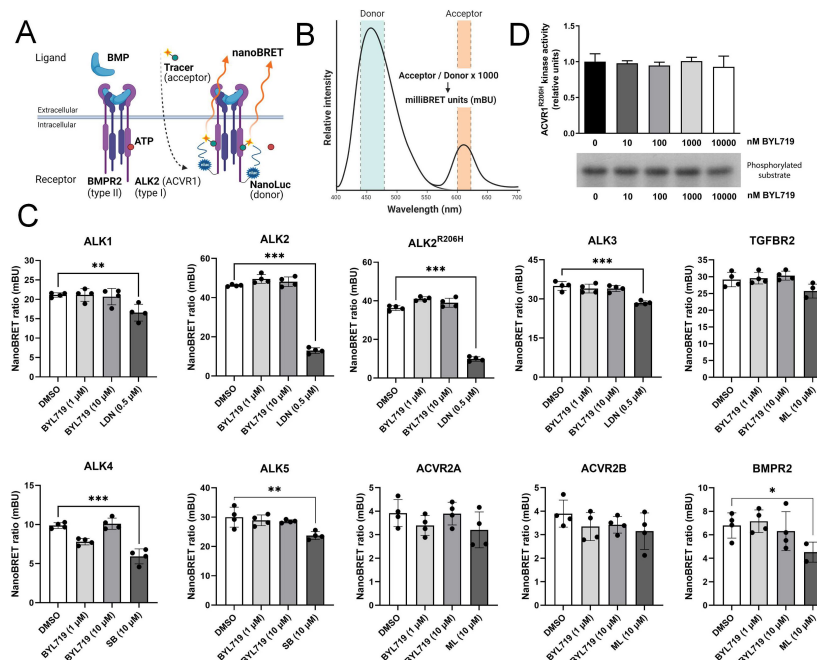


Figure 4.

(A) A schematic depiction of nano-bioluminescence resonance energy transfer (NanoBRET) target engagement assays. TGF- β receptors-Nanoluciferase fusion proteins are expressed in COS-1 cells and an ATP-like tracer analogue in close proximity with the Nanoluciferase donor allows energy transfer from the Nanoluciferase donor to the fluorescent tracer acceptor. An ATP analogue molecule (type I inhibitor) will compete with the fluorescent tracer, impairing close proximity donor-acceptor and reducing the nanoBRET ratio. (B) The nanoBRET emission spectra consist of the Nanoluciferase donor (460nm) and the fluorescent acceptor (610nm). The nanoBRET ratio is shown as milliBRET units (mBU) by dividing the acceptor emission by the donor emission times 1000. (C) NanoBRET target engagement analyses of ACVR1, ACVR1, ACVR1^{R206H}, BMPR1A, ACVR1B, TGFBR1, TGFBR2, ACVR2A, ACVR2B and BMPR2 testing 1 or 10 μ M BYL719 with $n=4$. As controls, LDN193189 (0.5 μ M), SB431542 (10 μ M), and ML347 (10 μ M) were used. Data are shown as mean $\pm$ SD. One-way ANOVA with Dunnett's multiple comparisons test. (D) Casein phosphorylation by ACVR1^{R206H} kinase. Phosphorylation was performed in the presence of ACVR1^{R206H} kinase and increasing concentrations of the PI3K α inhibitor BYL719. Quantification of kinase activity. Data shown as mean $\pm$ SD ($n=4$ independent experiments). One-way ANOVA with Tukey's multiple comparisons test. * $P<0.05$, ** $P<0.01$, *** $P<0.001$.

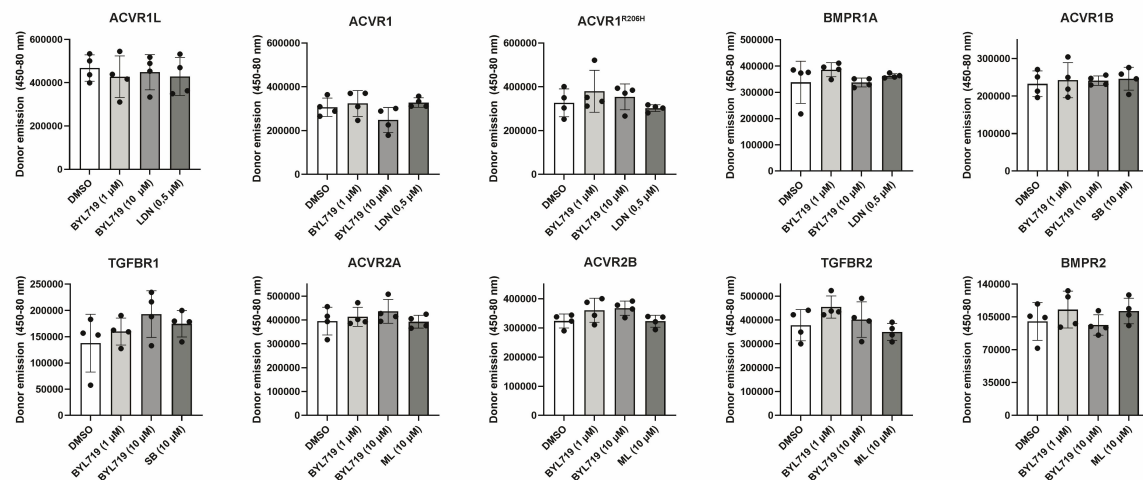


Figure 4-Figure supplement 1.

Quantification of the TGF- β receptor-Nanoluciferase protein expression levels measured by the raw donor emission (excitation at 450-480 nm).

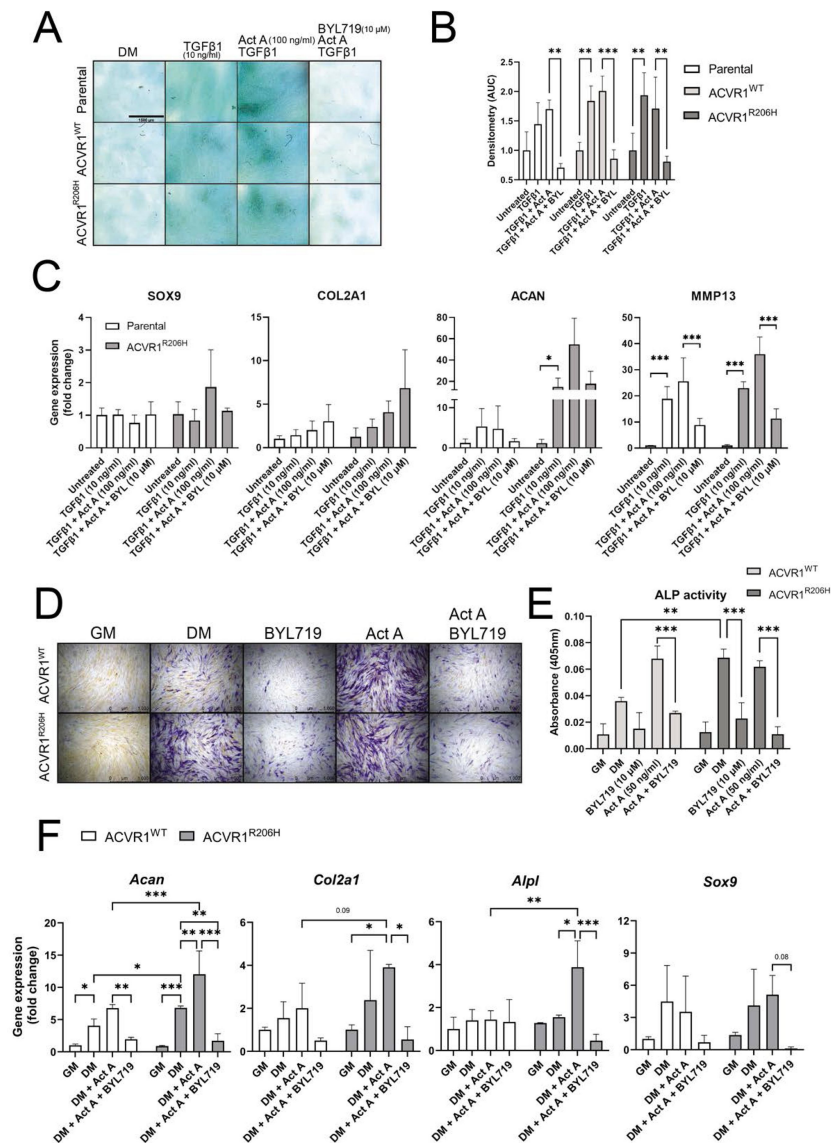


Figure 5.

Analyses of chondrogenic progenitor specification of parental human MSCs (hMSCs), hMSC-ACVR1^{WT} or hMSC-ACVR1^{R206H} treated with TGF-β1 (10 ng/ml) and Activin A (100 ng/ml), with and without BYL719 (10 μM) as indicated.

(A) Representative images of Alcian Blue staining of micromass cultures after 3 weeks of chondrogenic differentiation. Scale bar represents 1500 μm. (B) Densitometry analysis of the Alcian Blue staining in (A) (n=3 per group). (C) RT-qPCR results of *SOX9*, *COL2A1*, *ACAN* and *MMP13* after chondrogenic progenitor specification (n=4 per condition). (D) Representative images of the ALP staining after one week of chondrogenic progenitor specification. Scale bar represents 1000 μm. (E) Quantification of the ALP activity (n=3 per group). (F) RT-qPCR results of *Acan*, *Col2a1*, *Alpl* and *Sox9* after chondrogenic progenitor specification from *Acvr1*^{WT} or *ACVR1*^{R206H} MSCs treated with Activin A (100 ng/ml), with or without BYL719 (10 μM) in TGFβ1-containing differentiation media (DM) as indicated (n=3 per condition). Data are shown as mean ± SD. In all figure panels, significance is detailed between the indicated conditions. * P<0.05, ** P<0.01, *** P<0.001, two-way ANOVA with Tukey's multiple comparisons test.

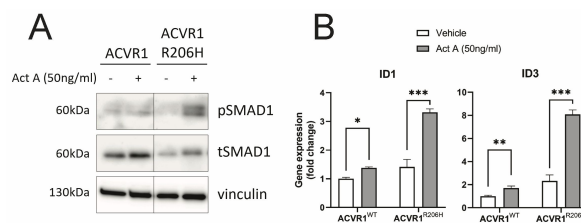


Figure 5-Figure supplement 1.

Human MSC-ACVR1^{R206H} are responsive upon Activin A (50 ng/ml) stimulation compared to hMSC-ACVR1^{WT} after overnight starvation.

(A) Western blot analysis detecting C-terminal phosphorylated SMAD1/5, total SMAD1 and vinculin. (B) RT-qPCR analysis of the BMP target genes ID1 and ID3 (n=3 per group). Data are shown as the mean $\pm$ SD. * P<0.05, ** P<0.01, *** P<0.001, by two-way ANOVA with Tukey's multiple comparison test.

To further identify the effects of BYL719, we performed bulk RNA sequencing (GSE237512) where we preincubated hMSCs-ACVR1^{WT} or hMSCs-ACVR1^{R206H} with or without BYL719 for 30 minutes, which was followed by stimulation with two high affinity ligands for ACVR1, Activin A (50ng/ml) or BMP6 (50ng/ml) for 1 hour (**Figure 6A** [↗](#)). Consistent with our results on osteogenic progenitor cells, gene ontology analysis of differentially expressed genes between cells expressing ACVR1^{WT} and ACVR1^{R206H} resulted in the robust regulation of relevant biological processes. Ossification (GO:0001503) and osteoblast differentiation (GO:0001649) were detected as two of the ten most significantly differentially regulated biological processes between these conditions (**Figure 6B** [↗](#)). Moreover, using gene ontology, we analyzed ossification and osteoblast differentiation biological processes in presence of ACVR1^{WT} or ACVR1^{R206H} receptor, with different ligands (BMP6 or Activin A), and with or without BYL719 inhibitor. The addition of BYL719 (1 μ M) resulted in a downregulation of these GO terms for all conditions except activin A stimulated ACVR1^{WT} cells (**Figure 6C** [↗](#)).

When comparing gene expressions profiles between ACVR1^{R206H} treated with Activin A, with or without BYL719, Gene Set Enrichment Analysis (GSEA) of our GO terms of interest showed negative normalized enrichment scores, consistent with our gene ontology analysis (**Figure 6D** [↗](#)). By looking at the leading edge we could investigate the most relevant genes within these GO terms, which included relevant genes of these pathways such as *PTGS2*, *MSX2*, *SOX9*, and *BMP2* (**Figure 6E** [↗](#)). Moreover, several inflammatory cytokine genes (e.g. *IL6* and *IL15*) and inflammatory signaling pathways (e.g. TNF and NF κ B), were downregulated by BYL719 (**Figure 6-Figure Supplement 1A and B** [↗](#)). Therefore, we hypothesized that, in addition to preventing the activation of osteogenic and chondrogenic differentiation pathways in progenitor cells, part of the mechanism by which BYL719 prevents HO *in vivo* may be due to the modulation of inflammatory responses.

PI3K α inhibition reduces the hyperinflammatory response in heterotopic ossification

To confirm whether BYL719 can modulate excessive inflammation at the injured sites, we examined its impact on monocytes, macrophages and mast cells *in vivo*. These cell populations are known to increase in number two to four days after injury (Convente et al., 2018 [↗](#); Sorkin et al., 2020 [↗](#)), which coincides with the effects observed following early administration of BYL719. Hindlimb muscle samples were collected at two, four, nine, sixteen and twenty-three days post-induction of HO with Cre viruses and cardiotoxin in *ACVR1^{Q207D/Δ}* mice (**Figure 7A** [↗](#)). The group of treated animals received BYL719 intermittent treatment starting one day after injury. After sixteen days, structures of mineralized bone became detectable in vehicle-treated mice (**Figure 7-Figure Supplement 1A and B** [↗](#)). The number of F4/80-positive monocytes/macrophages was high two days after injury and remained elevated in the HO lesions throughout the ossification process (**Figure 7B** [↗](#) and **C** [↗](#)). BYL719-treated animals showed an almost 50% reduction in the number of F4/80 positive cells at day two post-injury onwards. Consistent with previous studies, we observed an increased number of mast cells at injury sites. Their numbers peaked at four days and only dropped twenty-three days after injury in vehicle-treated mice (**Figure 7D** [↗](#) and **E** [↗](#)). In BYL719-treated mice, mast cells were increased up to day 9, but their number was completely normalized to preinjury levels already at day 16 (**Figure 7D** [↗](#) and **E** [↗](#)). This observation is consistent with the complete muscle regeneration observed in BYL719-treated mice compared to untreated mice, which are actively developing ossifications at that time (**Figure 7-Figure Supplement 2** [↗](#)). These results confirm a functionally relevant dual role of PI3K α inhibition, by preventing the differentiation of osteochondrogenic progenitor cells and also by reducing the local inflammatory response.

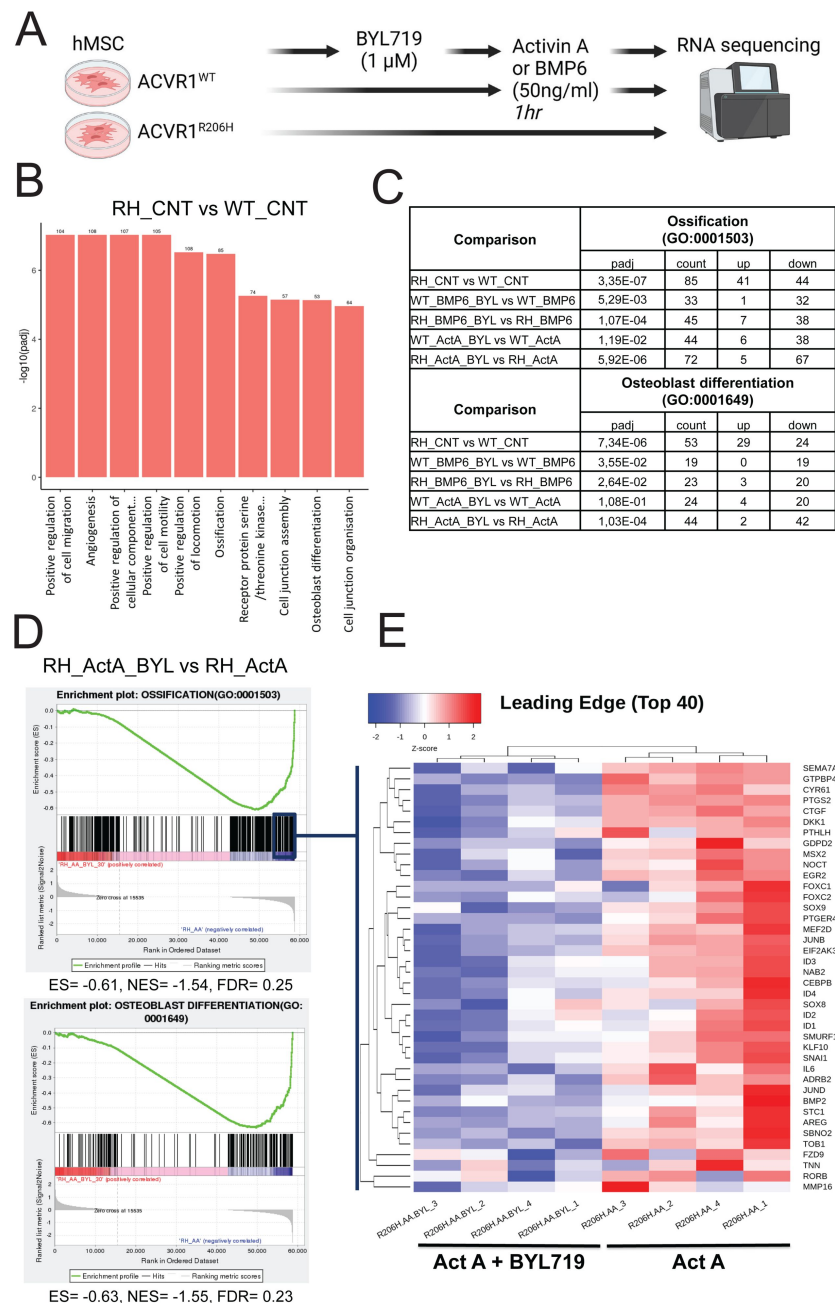


Figure 6.

(A) Schematic depiction of the experimental setup, involving bulk RNA sequencing of human MSC-ACVR1^{WT} and ACVR1^{R206H} upon overnight starvation, 30 minutes pre-treatment with or without 1 μM BYL719 and with or without 1 hour Activin A (50 ng/ml) or BMP6 (50ng/ml) stimulation. (B) The top 10 most significant gene ontology (GO) terms using all (up- and down-regulated) differentially expressed genes between cells expressing ACVR1^{WT} and ACVR1^{R206H} under control conditions. Ossification (GO:0001503) and osteoblast differentiation (GO:0001649) were detected within the top 10 of the differentially regulated biological processes. (C) Table of GO terms ossification and osteoblast differentiation upon GO enrichment analysis in all tested conditions ACVR1^{WT} or ACVR1^{R206H} cells, stimulated with BMP6 or Activin A and treated with BYL719. Statistical significance upon GO enrichment analysis, count of total differentially expressed genes (DEGs), and their classification as up- or down-regulated DEGs are detailed for each comparison. (D) Gene Set Enrichment Analysis (GSEA) with the groups RH_ActA_BYL vs RH_ActA, showing the enrichment plots of the gene ontology sets ossification (GO: 0001503) and osteoblast differentiation (GO:0001649). (E) A heatmap of the top 40 most relevant genes within the ossification GO geneset derived from the leading-edge subset of the enrichment plot (n=4 per group). Sample names are detailed as receptor (R206H) ligand (AA, Act A, Activin A)_inhibitor (BYL719, if present)_replicate#.

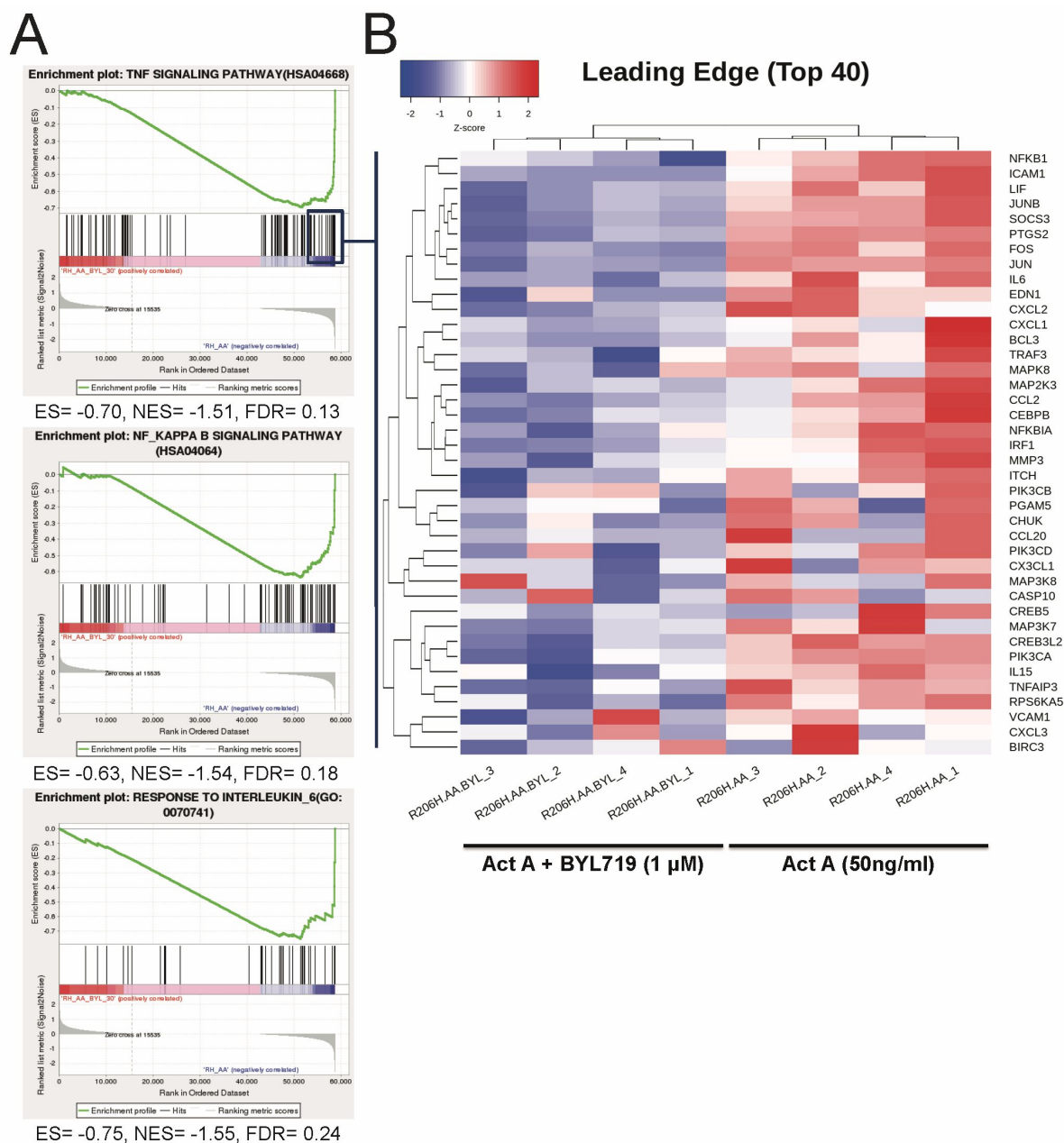


Figure 6-Figure supplement 1.

GSEA of bulk RNA sequencing comparing hMSC-ACVR1^{R206H} cells treated with BYL719 (1 μ M) with untreated controls, both stimulated with Activin A (50 ng/ml) (n=4 per group).

(A) Significant down-regulated enrichment plots of the KEGG gene set TNF signaling pathway (HSA04668), NF-kB signaling (HSA04064) and GO gene set response to interleukin-6 (GO:0070741). (B) The top 40 most relevant genes from the enrichment plot of the TNF signaling pathway as derived from the leading edge subset.

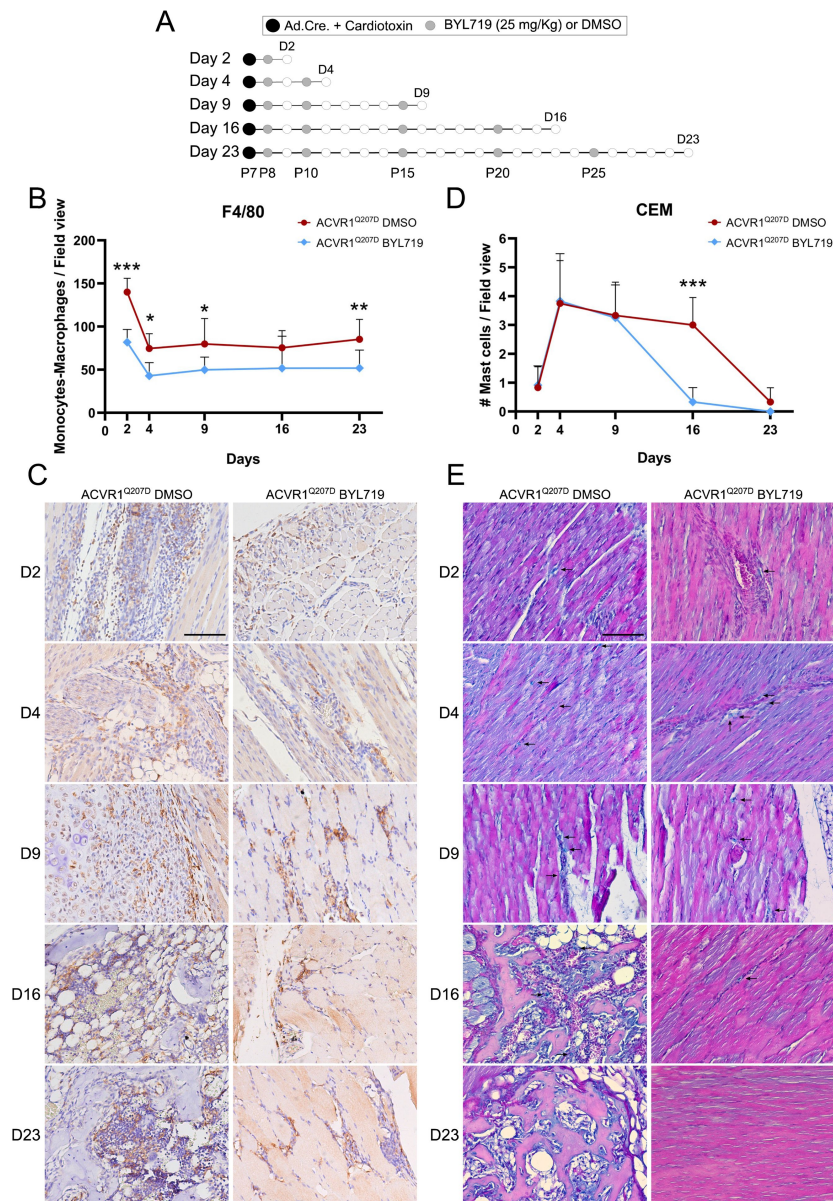


Figure 7.

(A) Heterotopic ossification was induced at P7 through the injection of Adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. Either DMSO (vehicle) or BYL719 (25 mg/kg) were injected following the scheme indicated with grey dots, starting at P8 but with a different duration P9 (2 days post-HO-induction), P11 (4 days post-HO-induction), P16 (9 days post-HO-induction), P23 (16 days post-HO-induction), and P30 (23 days post-HO-induction). (B) The quantification of F4/80-positive monocytes/macrophages per field view from immunohistochemistry staining, with 20x amplification as shown in the representative images depicted in [Figure 8C](#). Five images were randomly acquired per mice, from 2 mice per group, for a total of 10 quantified images per group. F4/80 positive cells were detected as cells with dark brown staining. Data are shown as mean $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, two-way ANOVA with Sidak's multiple comparisons test, comparing both groups on each day ($n = 10$). (C) Representative images from the quantified immunohistochemistry staining for F4/80 to detect monocytes/macrophages. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All images were acquired at 20x (scale bar 100 μ m). (D) Quantification of CEM-positive mast cells per field view from CEM staining, with 20x amplification as shown in the representative images depicted in [Figure 8E](#). Three images were obtained per mice, from 4 mice per group, for a total of 12 quantified images per group. Mast cells were detected as cells with bright blue staining. Data are shown as mean $\pm$ SD. *** $P < 0.001$, two-way ANOVA with Sidak's multiple comparisons test, comparing both groups in each day ($n = 12$). (E) C.E.M. staining was performed to detect mast cells, highlighted with black arrows. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All the images were obtained at 20x magnification, with a representative scale bar at 100 μ m.

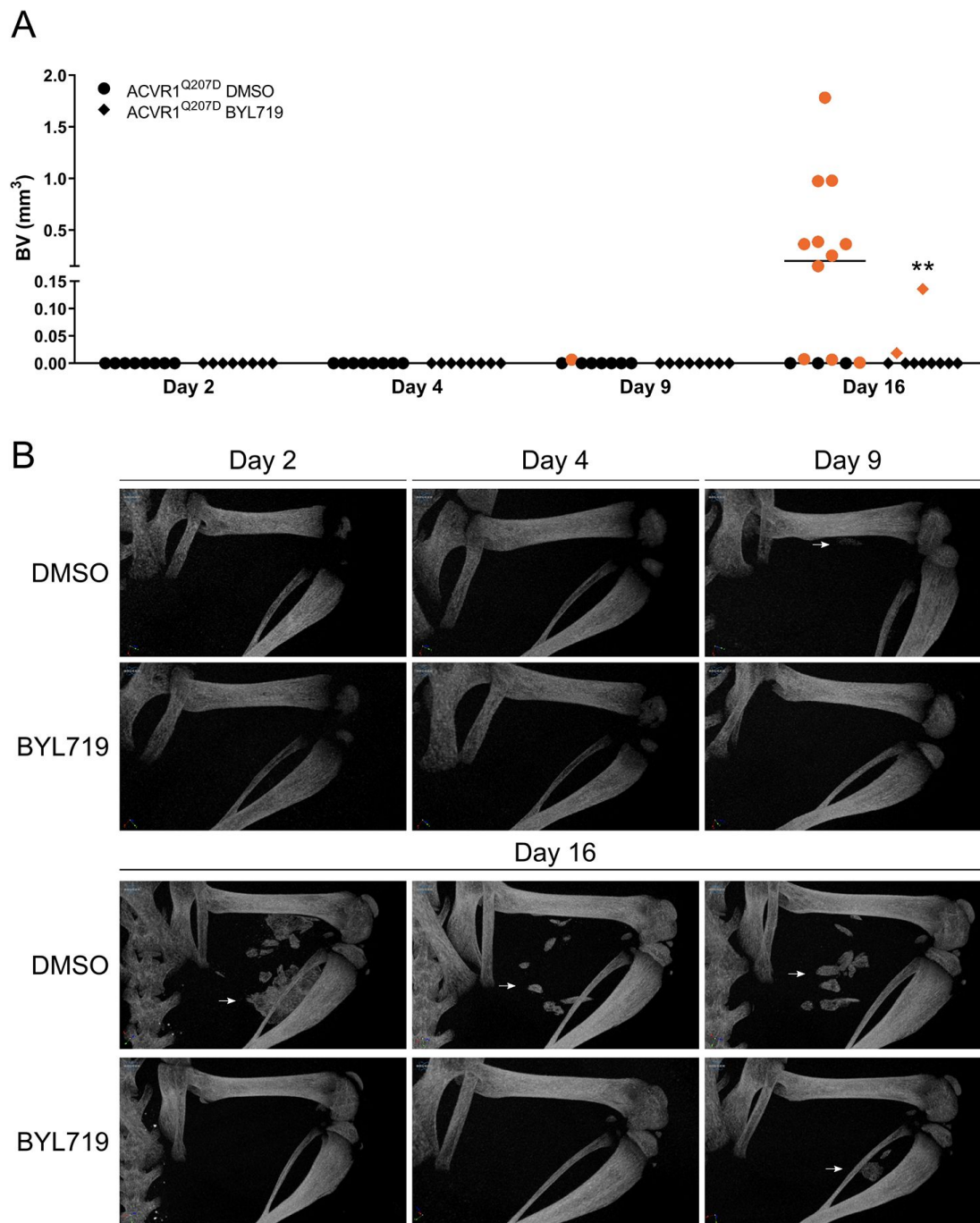


Figure 7-Figure supplement 1.

(A) Quantification of heterotopic ossifications bone volume (mm³) of each experimental group. group for each final time-point at day D-2, 4, 9, 16, and 23 post-HO-induction. Colored symbols indicate the presence of heterotopic ossifications. Black symbols indicate the absence of heterotopic ossifications. Circles indicate DMSO-treated mice, diamonds indicate BYL719-treated mice, as detailed in the legend. Data shown are of each individual mouse with group median. **P<0.01, two-way ANOVA with Sidak's multiple comparisons test. (B) Representative 3D microtomography images of the injected hindlimbs of mice from each experimental group. White arrows show heterotopic ossification.

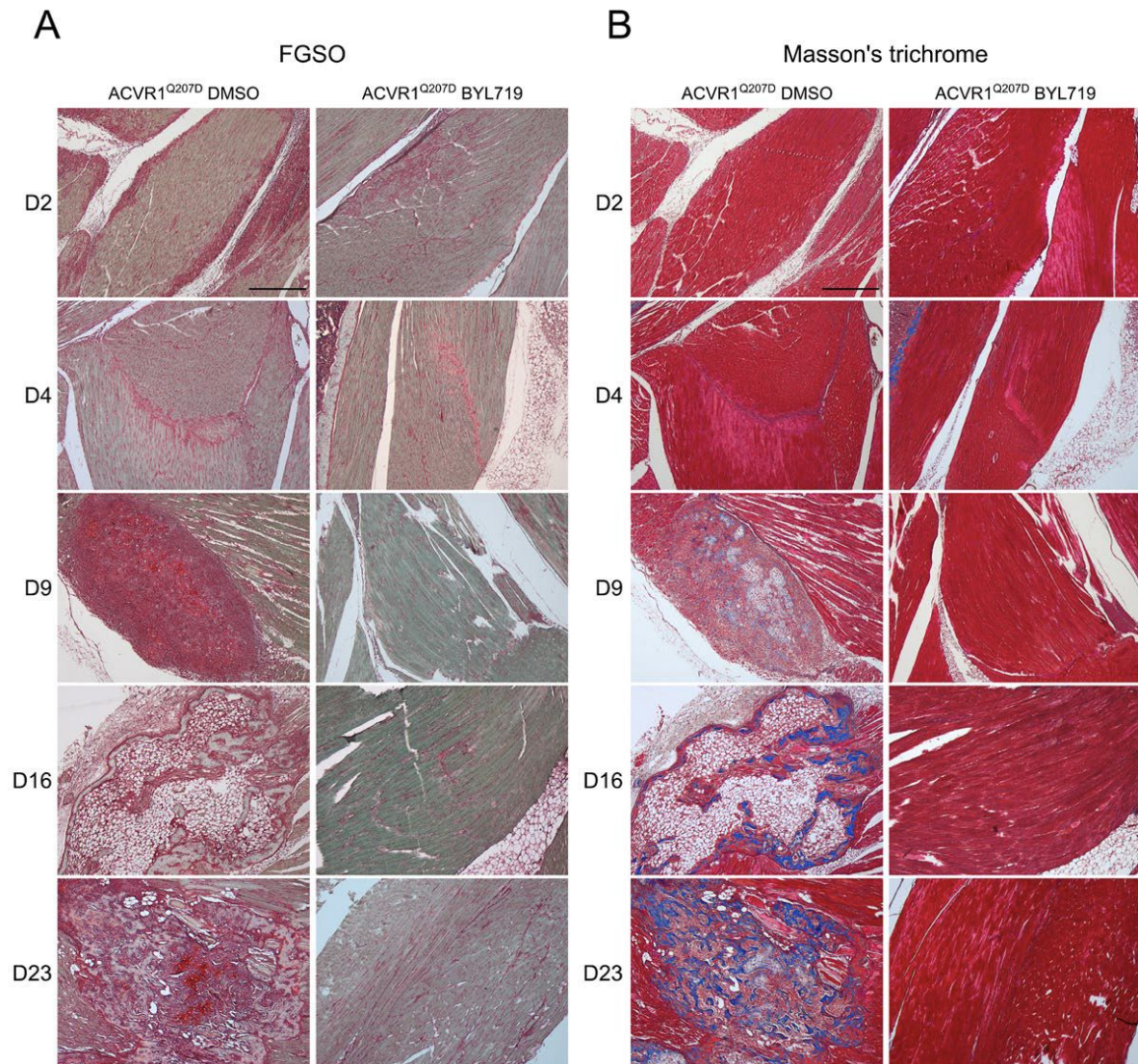


Figure 7-Figure supplement 2.

(A) Representative images of injected hindlimbs stained with Fast green/safranin O (FGSO) staining. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All images were obtained at 4x (scale bar at 500 μ m). (B) Representative images of injected hindlimbs stained with Masson's trichrome. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All images were acquired at 4x magnification (scale bar at 500 μ m).

PI3K α inhibition reduces proliferation, migration and inflammatory cytokine expression in monocytes, macrophages and mast cells

Next, we used several models of disease-relevant immune cells to study the effect of pharmacological PI3K α inhibition. Given the technical difficulties to transduce immune cells with lentiviral particles carrying ACVR1 R206H, we decided to partially recapitulate ACVR1 R206H activation with recombinant BMP6 and to test the effect of BYL719 in these conditions. Activation of ACVR1 signaling with BMP6 did not significantly modify the proliferation rate of any of the cell lines tested, that is, monocytes (THP1), macrophages (RAW264.7) and mast cells (HMC1) (**Figure 8A**, **B** and **C**). However, incubation with BYL719 *in vitro* strongly reduced their growth rate at 2 μ M and almost completely arrested their proliferation at 10 μ M. It is well known that monocytes are actively recruited from circulation to the ossification sites early after injury (Sorkin et al., 2020). We also examined the effects of BYL719 in proliferation in additional cell types involved in HO, such as myoblast and mesenchymal progenitors. BYL719 reduced the proliferation of myoblast and mesenchymal progenitors *in vitro* (**Figure 8**, Figure Supplement 1 A and B). However, the reduction in the proliferation did not reach the extent observed in monocytes or macrophages. We expanded our analysis by interrogating chemotactic migration of monocytes in transwell assays. Whereas monocytes actively migrated towards FBS as a chemotactic agent, the incubation of monocytes with BYL719 deeply diminished their migratory ability (**Figure 8D**). These results suggest that BYL719 is able to diminish monocyte recruitment, while reducing also the expansion of inflammatory cells at injury sites.

Upon tissue injury, a multitude of pro-inflammatory and anti-inflammatory cytokines are released at the site of injury. This triggers the recruitment and infiltration of various cell types, including immune cells and osteochondrogenic progenitors. We examined the expression of distinct cytokines upon ACVR1 stimulation with BMP6 in the presence of BYL719 in monocytic, macrophagic and mast cell lines. In monocytes, BYL719 inhibited the expression of the pro-inflammatory cytokine *TNF- α* and the expression of activin A (*INHBA*), whereas the expression of transforming growth factor *TGF- β 1* was increased by BMP6 and not modified by the addition BYL719 (**Figure 8E**). *CCL2* (MCP-1), a chemokine inducing the recruitment of monocytes and macrophages into inflamed tissues as well as the M2 polarization of macrophages (Sierra-Filardi et al., 2014), was significantly induced by BMP6 and partially reduced by BYL719 (**Figure 8E**). In macrophages, BYL719 reduced the expression of pan-macrophage markers *Adgre1* (F4/80), *Cd68* and the M1 macrophage marker *Cd80* (**Figure 8F**). Conversely, BYL719 increased the expression of the M2 macrophage marker *Il10*, regardless of the presence or absence of BMP6. In the human mast cell line HMC1, BYL719 inhibited the expression of the pro-inflammatory cytokine *IL6*, while the expression of *TGF- β 1*, *TNF α* , and activin A (*INHBA*) remained unchanged upon the addition of either BMP6 or BYL719 (**Figure 8G**). We also analyzed the effects of BYL719 on myogenic differentiation *in vitro*. Addition of 2 or 10 μ M BYL719 was able to abolish the expression of the myogenic markers *MyoD1* and myosin heavy-chain 1 (*Myh1*) while barely altering the expression of the myofibroblast marker α -SMA (*Acta2*) (**Figure 8-Figure Supplement 1C**). Similarly, BYL719 (at 2 or 10 μ M) was able to block formation of myotubes after 7 days of myogenic differentiation *in vitro* (**Figure 8-Figure Supplement 1D**). Altogether, these data suggest that PI3K α inhibition might reduce the expression of certain pro-inflammatory cytokines. Moreover, even though these results should be further validated in cells carrying mutated ACVR1, the reduction on activin A expression by BYL719 observed in inflammatory cells and mesenchymal progenitors could be relevant for HO in FOP.

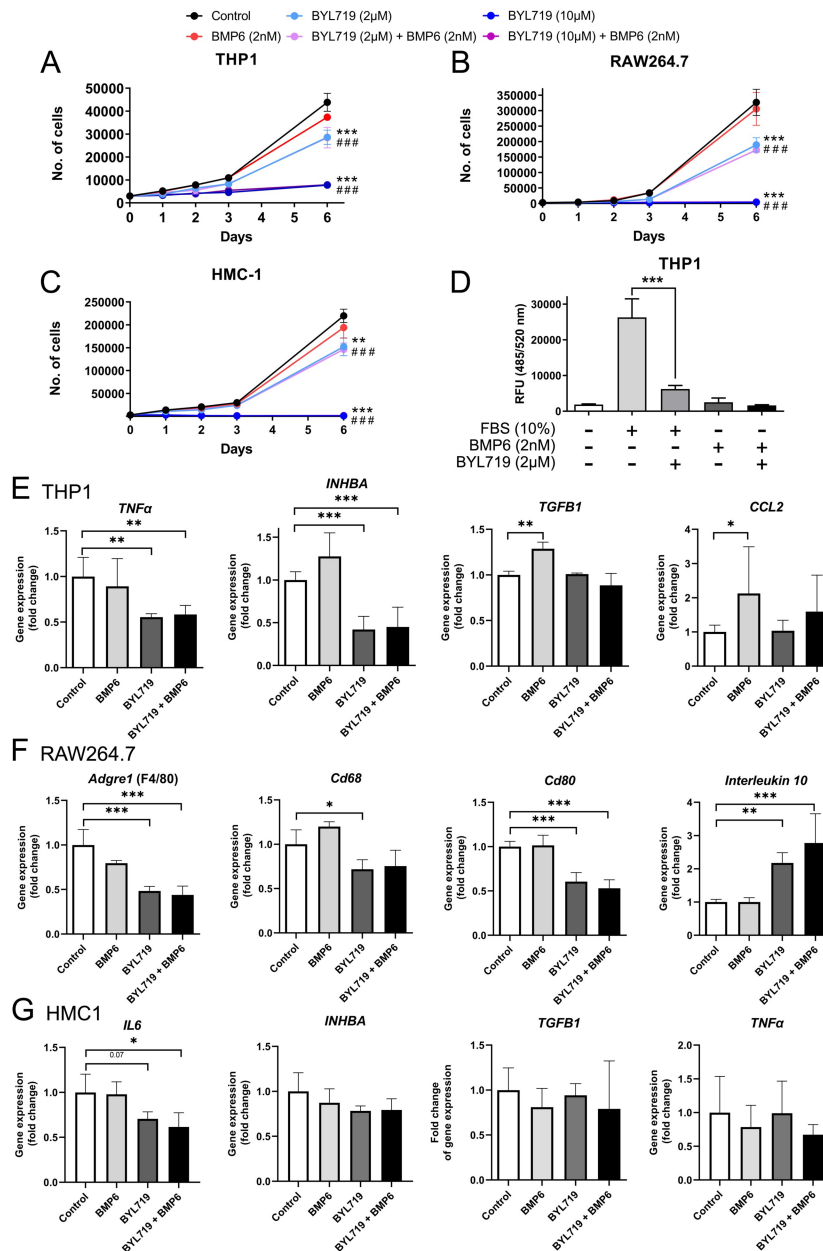


Figure 8.

(A, B, C) Proliferation assays of THP1 (A), RAW264.7 (B), and HMC-1 (C) cells. Cells were cultured for 6 days and in control conditions, with BMP6 (2nM) and/or BYL719 (2 or 10μM). Areas under the proliferation curves were compared by one-way ANOVA with Dunnett's multiple comparisons test against the control. (*) refers to significant differences between control and single treatments (BMP6 or BYL719). (#) refers to significant differences between control and combined treatments. Data shown as mean ± SD (n = 4 per group). ** or # P<0.01, *** or # # P<0.001. (D) Migration assay of THP1 cells with control conditions, or with FBS or BMP6 as chemotactic agents with or without BYL719 treatment. Data are shown as mean ± SD (n = 4 per group). ***P<0.001, two-way ANOVA with Tukey's multiple comparisons test. (E, F, G) Gene expression assays of THP1 (E), RAW264.7 (F) and HMC-1 (G) cells. Cells were treated for 48 hours with BYL719 (2μM) and/or BMP6 (2nM). Data are shown as mean ± SD (n = 4 per group). *P<0.05, **P<0.01, ***P<0.001, one-way ANOVA with Dunnett's multiple comparisons test, significance shown between control group and other groups.

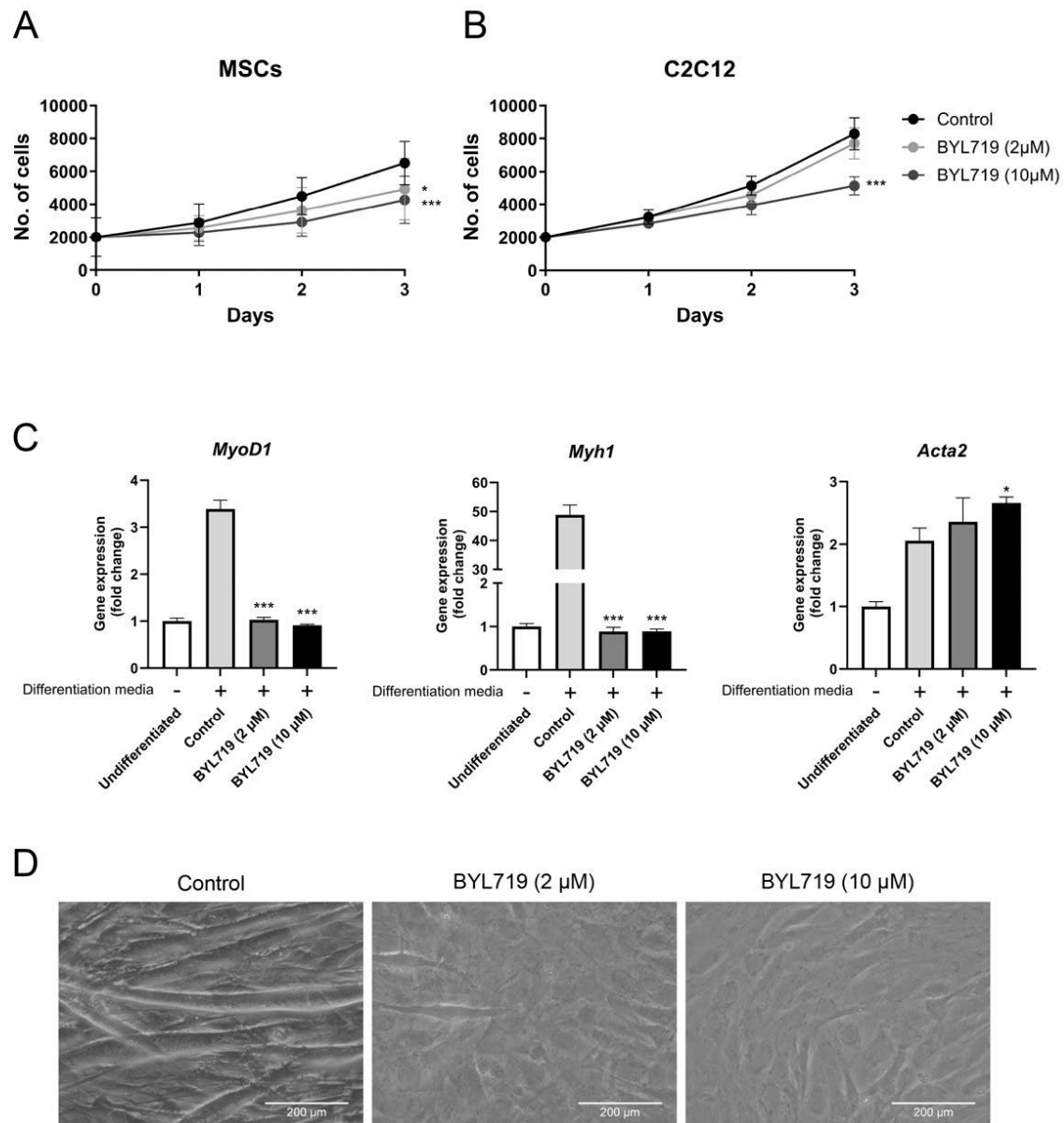


Figure 8-Figure Supplement 1.

(A and B) Proliferation assays of murine MSCs (A), and C2C12 cells (B).

Cells were cultured for 3 days and in control conditions, or with BYL719 (2 or 10 μM). Areas under the proliferation curves were compared by one-way ANOVA with Dunnett's multiple comparisons test against the control. (C) RT-qPCR analysis of the myogenic (*MyoD1* and *Myh1*) or myofibroblast (*Acta2*) markers after culturing C2C12 cells in myogenic differentiation media for 7 days in control conditions, or with BYL719 (2 or 10 μM) (n=3 per group). Data are shown as the mean ± SD. *P<0.05, **P<0.01, ***P<0.001, one-way ANOVA with Dunnett's multiple comparisons test. (D) Representative images of myotube formation of C2C12 cells in myogenic differentiation media for 7 days in control conditions, or with BYL719 (2 or 10 μM). Scale bar represents 200 μm.

Discussion

We previously showed that pharmacological PI3K α inhibition using BYL719 has the potential to suppress HO by increasing SMAD1/5 degradation, reducing transcriptional responsiveness to BMPs, and blocking non-canonical responses such as the activation of AKT/mTOR (Valer et al., 2019a [DOI](#)). In this manuscript, we confirmed these results through both pharmacological and genetic approaches, and further expanded the insights into the molecular and cellular mechanisms responsible for the therapeutic effect of PI3K inhibition. Here, we optimized the therapeutic window for BYL719 administration and found that the delayed administration of BYL719 effectively prevents HO. Moreover, *in vivo* simultaneous genetic inhibition of *p110 α* and the activation of mutated *ACVR1* at injury sites partially prevents HO, and we demonstrate that inhibition of PI3K α inhibits osteochondroprogenitor specification. We determined that BYL719 does not inhibit recombinant ACVR1^{R206H} kinase activity *in vitro* and does not behave as an ATP competitor inhibitor for none of the TGF- β receptor kinases. Finally, the administration of BYL719 prevented an exacerbated inflammatory response *in vivo*, possibly due to the effects observed on immune cell populations.

While FAPs play a key role in bone formation in different types of HO, inflammation is also closely associated with all types of HO (Barruet et al., 2018 [DOI](#); de Ruiter et al., 2021; Matsuo et al., 2019 [DOI](#)). For instance, the depletion of mast cells and macrophages has been shown to reduce bone volume in FOP mouse models (Convente et al., 2018 [DOI](#)). Moreover, Activin A, TGF- β and other cytokines secreted by FAPs, monocytes, macrophages and mast cells are essential for FOP and non-genetic HO (Alessi Wolken et al., 2017 [DOI](#); Hatsell et al., 2015 [DOI](#); Lees-Shepard et al., 2018 [DOI](#); Patel et al., 2022 [DOI](#); Sorkin et al., 2020 [DOI](#); Upadhyay et al., 2017 [DOI](#)). In addition, ACVR1 antibodies can activate ACVR1^{R206H} even in the absence of Activin A, but muscular trauma is still required to induce HO (Aykul et al., 2022 [DOI](#); Lees-Shepard et al., 2022 [DOI](#)), indicating the involvement of additional immune factors. Our study finds that BYL719, a PI3K α inhibitor, reduces the expression of certain pro-inflammatory cytokines, effectively blocks the proliferation of monocytes, macrophages and mast cells and reduces the migratory potential of monocytes. This likely contributes to a decreased number of monocytes and macrophages at injury sites and throughout the *in vivo* ossification process. It is known that excessive pro-inflammatory cytokine expression, including activin A, by monocytes and macrophages is observed in all types of HO and is induced by ACVR1^{R206H} (Matsuo et al., 2021 [DOI](#)). Non-canonical signaling pathways (mTOR, p38, TAK1, NF- κ B) are also affected by ACVR1^{R206H} in immune cell types (Barruet et al., 2018 [DOI](#); Hwang et al., 2022 [DOI](#)). M1-type macrophages initiate acute inflammatory responses but later transition to M2-type macrophages associated with anti-inflammatory and reparative functions. Therefore, modulating macrophage phenotype towards regeneration may dampen HO (Sorkin et al., 2020 [DOI](#)). BYL719 reduces monocyte/macrophage numbers, migration, and pro-inflammatory cytokine expression, potentially altering their polarization. It also reduces activin A expression and promotes a shift towards the M2 phenotype. BYL719 can help to reduce the hyper-inflammatory state in HO, inhibit FAP expansion, and favor myogenic regeneration of muscle tissue (Stanley et al., 2022 [DOI](#)).

Our observation that the delayed administration of BYL719 is still effective has pathophysiological and therapeutic derivatives. Muscle injury in mice induces sequential changes in HO progression: early immune cell infiltration (days 1-3), a fibroproliferative and late inflammatory phase, with an expansion of FAPs and a spectrum of monocytes, macrophages and mast cells as highly secretory cells (days 3-7), followed by chondrogenesis (day 7-14) and osteogenesis (days 14-23) (Hwang et al., 2022 [DOI](#)). Starting BYL719 administration 3 days post-injury is still fully therapeutically effective and even partially at 7 days post-injury. In addition, whereas untreated mice developed cartilage and bony lesions by days 9 and 16 post-injury, respectively, BYL719-treated animals fully regenerated muscle by day 16 post-injury. Altogether, this evidence suggests that the major effects of BYL719 occur between 3 and 16 days post-injury. We found that BYL719 completely blocks

chondrogenesis in cultured hMSCs. In addition, in the histological analyses of these BYL719-treated mice that do not develop HO, we cannot find any sign of cartilage formation on either the 9th, 16th or 23rd days post-injury, which confirms that BYL719 is also able to block chondrogenesis in mice *in vivo*. This suggests that intervention with BYL719 over a temporal window after a HO flare might be sufficient to inhibit endochondral ossification. In addition, although surgical resection is not recommended in FOP patients and has a risk of recurrence in resected non-genetic HO, it could be envisaged using as preventive therapy in subjects undergoing surgeries to remove ectopic bone.

Treatment options for HO and FOP are currently limited, primarily consisting of anti-inflammatories such as corticosteroids and NSAIDs. Promising pharmacological treatments have progressed to II/III clinical trials. Palovarotene, a retinoic acid receptor- γ agonist, has been approved in Canada and U.S.A. for patients over 10 years old (NCT03312634). However, a previous clinical trial with palovarotene was paused for children due to concerns about premature growth plate closure, which is debated in FOP mice (Chakkalakal et al., 2016 [↗](#); Rosen et al., 2018 [↗](#)) (Chakkalakal et al., 2016 [↗](#); Rosen et al., 2018 [↗](#)). Antibodies targeting activin A (GaretoSmab) showed effectiveness, but serious adverse effects led to a temporary hold on the trial (NCT03188666). The outcome of a clinical trial with Rapamycin has not been publicly disclosed and a case report indicated limited benefits in classical FOP patients (UMIN000028429) (Kaplan et al., 2018 [↗](#)). BYL719 (Alpelisib/Piqray®) was recently licensed in combination with hormone therapy for a PIK3CA-mutated breast cancer (Cardoso et al., 2020 [↗](#)). BYL719 has been shown to be also clinically effective in adult patients with PIK3CA-related overgrowth syndrome (PROS) and children under 1 year of age (Madsen and Semple, 2022 [↗](#); Morin et al., 2022 [↗](#)). In both scenarios, BYL719 had no major adverse effect when patients were treated for more than one year (in adults at a dose of 300mg daily for oncological or PROS purposes and at a dose of 25 mg daily in infants with PROS) (Morin et al., 2022 [↗](#)). Thus, implementing BYL719 for treatment of HO, at least during a restricted temporal window (i.e. surgery to remove ectopic bone in HO, or during flare-ups in FOP individuals), might be a valid therapeutic option for FOP patients.

Materials and methods

Murine bone marrow mesenchymal stem cells isolation

Murine bone marrow-derived mesenchymal stem cells (BM-MSCs) were isolated from 6-8 weeks-old C57BL6/J mice for the floxed p110 α allele (Beatriz Gámez et al., 2016 [↗](#)) as previously described (Soleimani and Nadri, 2009 [↗](#)). We also isolated BM-MSCs from UBC-CRE-ERT2/ACVR1^{R206H}*fl/wt* (a gift from Dr. Dan Perrien and IFOPA). This conditional ACVR1^{R206H} construct was knocked-in to the endogenous *Acvr1* gene immediately following intron 4. After 4=OH tamoxifen addition, CRE activity excises murine *Acvr1* exons 5-10 and induce expression of the corresponding exons of human ACVR1^{R206H} and an eGFP marker. In both cases, mice were euthanized and femurs were dissected and stored in DMEM with 100 U/ml penicillin/streptomycin (P/S). Soft tissues were cleaned and femur ends were cut under sterile conditions. Bone marrow was flushed with media using a 27-gauge needle. The resulting cell suspension was filtered through a 70- μ m cell strainer and seeded in a 100-mm cell culture dish. Non-adherent cells were discarded after 3 hours. Media was slowly replaced every 12 hours for up to 72 hours. Then, media was replaced every 2 days until the culture reached 70% confluence. Then, cells were lifted by incubation with 0.25% trypsin/0.02% EDTA for 5 minutes at room temperature. Lifted cells were cultured and expanded. MSCs from UBC-CRE-ERT2/ACVR1^{R206H}*fl/wt* mice were treated with 4-hydroxytamoxifen to induce Cre recombination.

Production of human mesenchymal stem cells (hMSCs) expressing ACVR1

For chondrogenic differentiation and RNA sequencing, bone marrow-derived human MSCs were transduced by lentiviral delivery encoding ACVR1^{WT} or ACVR1^{R206H} (Van Dinther et al., 2010 [DOI](#)). For NanoBRET target engagement kinase assays, were transduced with ACVR1^{WT}-nanoluc or ACVR1^{R206H}-nanoluc. The nanoluc fusion inserts (Promega, NV2341/NV2381) were subcloned into a PLV-CMV-IRES vector. First, an eGFP was amplified with PCR containing PstI, SalI and XbaI restriction sites and subcloned in the PLV using PstI and XbaI. Next, the ACVR1-nanoluc inserts were subcloned in the PLV by restriction of PstI and SalI, exchanging the eGFP. Subsequently, lentiviral particles were produced using HEK293t cells and cells were transduced.

Cell culture

Murine BM-MSCs were cultured in DMEM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate, and 100 U/ml P/S and incubated at 37°C with 5% CO₂. The human monocyte cell line THP1 was cultured in RPMI-1640 supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate, 100 U/ml P/S, 0.1mM NEAA, 10mM HEPES and 50μM 2-mercaptoethanol. Mouse macrophage cell line RAW264.7 was cultured in DMEM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate and 100 U/ml P/S. The human mast cell line HMC1 was cultured in IMDM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate and 100 U/ml P/S. The hMSCs were cultured in αMEM supplemented with 10% FBS, 0.1 mM ascorbic acid, 1 ng/ml human bFGF and 100 U/ml P/S (Growth Medium; GM). C2C12 cells were cultured in DMEM supplemented with 20% FBS, 2 mM glutamine, 1 mM sodium pyruvate, and 100 U/ml P/S and incubated at 37°C with 5% CO₂. For C2C12 differentiation, cells were cultured for seven days in the same media with 5% horse serum.

Proliferation and chemotactic migration assays

For proliferation assays, cells were seeded in 12-well culture plates in correspondent culture medium with 2 nM BMP6 and/or 2 μM or 10 μM BYL719 (Chemietek). Photos of several fields of each well were taken at 24, 48, 72 and 144 hours using a Leica DM-IRB inverted microscope. Cell counting was performed through ImageJ software.

The chemotactic migration assay of THP1 was performed through the InnoCyte Monocyte Migration Assay (Merck Millipore). Cells were serum-starved for 3h prior to the experiment and, when needed, treated with 2 μM BYL719 for 30 minutes before the experiment started. Then, 10⁵ cells/100 μL were added to each upper chamber. Lower chambers were charged with 10% FBS-supplemented RPMI-1640 (positive control), FBS-depleted RPMI-1640 (negative control) or 2 nM BMP6 (R&D), respectively. Cells migrated for 2h and were then stained with calcein-AM and pipetted into a 96-well conical bottom plate (Sigma-Aldrich). A plate reader was used to measure fluorescence from the top of the plate at an excitation wavelength of 485 nm and an emission wavelength of 520 nm. Unless otherwise stated, BYL719 at 2 or 10 μM (Chemietek) and 2 nM BMP6 (R&D) was used for 6 days (proliferation) or 4 hours (migration) in HMC1, THP1 and RAW264.7 cells.

Retroviral transduction and gene expression analysis

BM-MSCs were infected with mock virus (pMSCV) or with viruses expressing wild type *Acvr1* (WT), *Acvr1*^{R206H} (RH) and/or pMSCV-Cre. Plasmids with WT and RH *Acvr1* forms were kindly provided by Dr. Petra Seemann and were subcloned into pMSCV vector. *Acvr1* and p110α expression levels were analyzed by qRT-PCR. Cells were treated with BYL719 (2μM) and/or activin A (2nM) for 48 hours in complete media without FBS. THP1, RAW264.7, and HMC-1 cell lines were treated in the media indicated for each cell line with BYL719 (2μM) and/or BMP6 (2nM) for 48 hours.

Total RNA for RT-PCR was extracted from all cellular models using TRIsure reagent. At least 2 µg of purified RNA was reverse-transcribed using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). Quantitative PCRs were carried out on ABI Prism 7900 HT Fast Real-Time PCR System with Taqman 5'-nuclease probe method and SensiFAST Probe Hi-ROX Mix. All transcripts were normalized using *Tbp* as an endogenous control.

NanoBRET target engagement assays

COS-1 cells were cultured in DMEM (Gibco, 11965092) supplemented with 10% FBS and 100 U/ml P/S. These cells were transfected in 70% confluent 6-wells TC-treated plates with 2 µg TGFβ receptor-Nanoluciferase constructs (constructs were provided by Promega and Promega R&D) using a 1 µg DNA: 2 µl PEI (1 mg/ml) ratio. The transfected cells were reseeded in white 384-wells TC-treated assay plates one day before the nanoBRET readout in DMEM supplemented with 1% FBS in a quantity of 15×10^3 cells in 40 µl per well. Two hours before measurement, the dedicated tracer (Promega) and the test compounds were incubated to the cells at 37°C. Immediately after intracellular TE Nano-Glo Substrate/Inhibitor (Promega, N2160) addition, the wells were measured by 450-80BP and 620-10BP filters using the ClarioSTAR (BMG Labtech). Subsequent nanoBRET ratios were measured by the formula: acceptor emission (620-10 nm)/donor emission (450-80 nm) * 1000 (milliBRET units, mBU).

Construct	Cat. #	Tracer (conc.)	Cat. #	Compounds (conc.)
<i>ACVRL1-nluc</i>	NV2391	K-11 (0.31 µM)	N2652	BYL719 (1 and 10 µM) LDN193189 (0,5 µM)
<i>ACVR1-nluc</i>	NV2341	K-11 (0.16 µM)	N2652	BYL719 (1 and 10 µM) LDN193189 (0,5 µM)
<i>ACVR1 R206H-nluc</i>	NV2381	K-11 (0.16 µM)	N2652	BYL719 (1 and 10 µM)
<i>nluc</i>				LDN193189 (0,5 µM)
<i>BMPR1A-nluc</i>	NV2471	K-11 (0.63 µM)	N2652	BYL719 (1 and 10 µM) LDN193189 (0,5 µM)
<i>ACVR1B-nluc</i>	NV1021	K-5 (2 µM)	N2482	BYL719 (1 and 10 µM) SB431542 (10 µM)
<i>TGFBR1-nluc</i>	Promega R&D	K-14 (0.5 µM)	Promega R&D	BYL719 (1 and 10 µM) SB431542 (10 µM)
<i>ACVR2A-nluc</i>	Promega R&D	PBI6948 (1 µM)	Promega R&D	BYL719 (1 and 10 µM) ML347 (10 µM)
<i>ACVR2B-nluc</i>	Promega R&D	PBI6948 (1 µM)	Promega R&D	BYL719 (1 and 10 µM) ML347 (10 µM)
<i>TGFBR2-nluc</i>	Promega R&D	K-11 (0.32 µM)	N2652	BYL719 (1 and 10 µM) ML347 (10 µM)
<i>BMPR2-nluc</i>	Promega R&D	PBI7394 (0.5 µM)	Promega R&D	BYL719 (1 and 10 µM) ML347 (10 µM)

Chondrogenic progenitor differentiation

MSCs were cultured in αMEM supplemented with 10% FBS, 1 ng/ml human bFGF and 100 U/ml P/S (Growth Medium; GM). Chondrogenic differentiation medium (DM) consisted of DMEM/F12 supplemented with 1% Insulin-Transferrin-Selenium, 170 µM Ascorbic Acid, 0.1 µM Dexamethasone, 350 µM L-proline, 1 mM Sodium Pyruvate, 0.15% glucose, 1% FBS and 100 U/ml P/S. During differentiation, the medium was refreshed twice a week for the duration of the experiment. For the Alcian Blue experiments, micromass cultures were made by carefully seeding 3×10^5 cells per 10 µl droplets in a 24-wells plate and incubated for 2 hours at 37°C prior to GM

addition. Chondrogenic progenitor differentiation was started 1 day after micromass seeding by the addition of DM. Dependent on the condition, we supplemented the DM with TGF- β 1 (10 ng/ml) or Activin A (100 ng/ml), and treated with BYL719 (10 μ M) or DMSO vehicle. For the ALP staining, the hMSC-ACVR1 lines were seeded at 2×10^4 cells per well in 48-wells plates and differentiated for 7 and 11 days.

Alkaline phosphatase (ALP) staining, ALP activity assay and Alcian Blue staining

For the ALP staining, the cells were washed with PBS, fixed using 3.7% formalin for 5 minutes at RT, washed twice with PBS and stained in ALP solution containing 2 mg Naphthol AS-MX, 6 mg Fast Blue, 5 ml 0.2M Tris (pH 8.9), 100 μ l MgSO_4 and dH_2O up to 10 ml. Images were acquired using the Leica DMI8 with a 10x magnification. To measure ALP activity, ALP lysates were obtained by washing twice with PBS, freezing at -80°C for 1 hour, and lysing on ice for 1 hour using 100 μ l ALP buffer (100 μ M MgCl_2 , 10 μ M ZnCl_2 , 10 mM Glycine (pH 10.5)) plus 0.1% Triton X-100 per well. ALP activity was quantified by adding 20 μ l lysate and 80 μ l 6 mM PNPP in ALP buffer to a clear 96-well plate incubated at RT until the samples turned yellow. Next, the samples were measured at an absorbance of 405 nm.

The micromass pellets were stained with Alcian Blue as described before ([Sánchez-Duffhues et al., 2019](#)). Images were acquired using the Leica DMI8 with 5x magnification. Densitometry analysis of the whole well (from total plate scan) was performed using ImageJ (v1.53t).

Bulk RNA sequencing and analyses

The hMSCs-ACVR1^{WT} and hMSCs-ACVR1^{R206H} were seeded at 100% confluency and serum starved overnight with DMEM without supplements. Afterwards, the cells were pre-treated with or without 1 μ M of BYL719 in starvation medium for 30 minutes and stimulated with or without 50 ng/ml Activin A or 50 ng/ml BMP6 for 1 hour. RNA was isolated using the ReliaPrep RNA Miniprep Systems (Promega) and sequenced using the Illumina NovaSeq 6000 (Illumina) platform. The reads were mapped using the Hisat2 (v2.0.5) package and the total and normalized counts were measured using featureCounts (v1.5.0-p3). Differential expression analysis was performed using the DESeq2 R package (v1.20.0), in which the p-values were adjusted following the Benjamini and Hochberg approach. Genes with an adjusted p-value < 0,05 were considered differentially expressed. Subsequent Gene Ontology analysis was performed using the clusterProfiler R package and the GO terms were considered significant if the p-values < 0,05. Gene Set Enrichment Analysis was performed using the GSEA analysis tool ([gsea-msigdb.org](#)). Heatmaps were generated using heatmap2 function within the R gplots package in Galaxy ([usegalaxy.eu](#)). The datasets are publicly available (see GSE237512).

Western blot

Cells were lysed using RIPA lysis buffer and quantified using the Pierce BCA Protein assay (Thermo) following standard protocols. The protein samples were loaded using β -ME containing sample buffer and run using SDS-PAGE and transferred to PVDF membranes. Primary antibodies against SMAD1 (Cell Signaling, 6944), p-SMAD1/5 (Cell Signaling, 9516), and vinculin (Sigma, V9131) were used at 1:1000 dilution in 5% BSA in TBST. Binding was detected with HRP-conjugated secondary antibodies and visualized by Brightfield ECL (Thomas Scientific) on the ChemiDoc (BioRad).

In vitro recombinant kinase activity assay

Fifty ng of recombinant ACVR1^{R206H} (ab167922, Abcam) was diluted in 30 μ l of 25mM Tris/HCl (pH7.5), 10mM MgCl_2 , 10mM MnCl_2 and 2mM DTT. The reaction was started by the addition of 5 μ M ATP-Mg, 0.05 μ Ci/ μ l ATP [γ - ^{32}P] and 0.5mg/ml of dephosphorylated casein (Sigma) and performed

for 30 min at 30°C. The reaction was stopped and samples were processed by SDS-PAGE. After drying the gel, phosphorylation was determined by autoradiography.

Heterotopic ossification mouse model

To study heterotopic ossification *in vivo*, we used the Cre-inducible constitutively active ACVR1^{Q207D} (CAG-Z-EGFP-caALK2) mouse model as previously described (Fukuda et al., 2006 [↗](#); Shimono et al., 2011 [↗](#); Yu et al., 2008 [↗](#)). Mice ACVR1Q207D^{fl/fl} p110α^{fl/fl} were obtained by crossing the detailed homozygous ACVR1Q207D^{fl/fl} with the p110α^{fl/fl} homozygous mutant mice carrying loxP sites flanking exons 18 and 19 of the p110α alleles (Beatriz Gámez et al., 2016 [↗](#)). Heterozygous mice were crossed until ACVR1Q207D^{fl/fl} p110α^{fl/fl} was obtained.

To induce heterotopic ossification in P7 ACVR1^{Q207D} mice, 1×10⁸ pfu of adenovirus-Cre (Ad-CMV-Cre, Viral Vector Production Unit, UAB) and 0.3 μg of cardiotoxin in 10 μl 0.9% NaCl volume were injected into the left hindlimb. Control groups had the same procedure without Ad-Cre in the injection. On P8, mice started receiving either placebo (intraperitoneal administration (i.p.) of DMSO) or BYL719 (i.p. of 25 mg/kg), both diluted in 0.5% carboxymethylcellulose sodium. Mice were housed under controlled conditions (12-hour light/12-hour dark cycle, 21°C, 55% humidity) and fed *ad libitum* with water and a 14% protein diet (Teklad2014, Envigo). Mice were regularly weighted over the whole period. Heterotopic ossification induction was performed blinded to mouse and group identity. All procedures were approved by the Ethics Committee for Animal Experimentation of the Generalitat de Catalunya.

Micro-computed tomography analysis

The caudal half of mice was collected and fixed in 4% paraformaldehyde (PFA) for 48 hours at 4°C. Samples were conserved in PBS and high resolution images were acquired using a computerized microtomography imaging system (Skyscan 1076, Bruker microCT), in accordance with the recommendations of the American Society of Bone and Mineral Research (ASBMR). Samples were scanned in air at 50 kV and 200 μA with an exposure time of 800 ms, using a 1 mm aluminum filter and an isotropic voxel size of 9 μm. Two dimensional images were acquired every 1° for 180° rotation and subsequently reconstructed, analyzed for bone parameters, and visualized by NRecon v1.6, CT-Analyser v1.13 and CTVox v3.3 programs (Bruker), respectively. For heterotopic ossification, manual VOIs comprising heterotopic ossifications were employed and a binary threshold was established at 25-255.

Histology and immunofluorescence

Whole legs were fixed in 4% PFA for 48 hours at 4°C, decalcified in 16% EDTA pH 7.4 for 6 weeks, and embedded in paraffin. Five μm sections were cut and stained with Fast Green/Safranin O, or Masson's Trichrome. Images were obtained with brightfield Eclipse E800 (Nikon). To detect monocyte/macrophage cells, paraffined slides were immunostained with Anti-F4/80 antibody [SP115] (ab111101). Briefly, after deparaffination and rehydration, an antigen retrieval step with citrate buffer at pH6 in a decloaking chamber was applied to the slides. Slides were cooled down and rinsed with 1x TBS and endogenous peroxidases were blocked with 70% methanol, 28% distilled water, and 2% hydrogen peroxide for 5 minutes at room temperature. After a blocking step, F4/80 primary antibody incubation at concentration 1:100 overnight at 4°C was applied. Envision Dual Link conjugated with HRP was applied for 40 minutes at room temperature, and detection was performed with diaminobenzidine (DAB) for 3 minutes. Nuclear staining with Lilly's hematoxylin was applied to the slides for 30 seconds. Images of stained slides were obtained with brightfield Eclipse E800 (Nikon). Images were randomly acquired with 20x magnification from the region of interest (injected muscle, with or without visible heterotopic ossification). Five images were obtained per mice, from 2 mice per group, for a total of 10 quantified images per group. F4/80 positive cells were detected as cells with dark brown staining. To detect mast cells, paraffined slides were stained with C.E.M. staining using the Eosinophil-Mast cell staining kit

(ab150665, Abcam). The recommended staining procedure was applied to the deparaffinized sections. Images of stained slides were obtained with brightfield Eclipse E800 (Nikon). Images were randomly acquired with 20x magnification from the region of interest (injected muscle, with or without visible heterotopic ossification). Three images were obtained per mice, from 4 mice per group, for a total of 12 quantified images per group. Mast cells were detected as cells with bright blue staining.

For immunofluorescence assays, five μm sections of each time point and condition were deparaffinized, processed with antigen retrieval and permeabilization steps, and stained with wheat germ agglutinin (1:250 dilution, ThermoFisher Scientific), followed by incubation using PDGFRA antibody (1:250 dilution Cell Signalling #3174), Alexa555 secondary antibody, and DAPI staining. Images were randomly acquired with 20x magnification from the region of interest with a Carl Zeiss Axio Imager M2 Apotome microscope. Five images were obtained per mice, from 4 mice per group, for a total of 20 quantified images per group.

Statistical analysis

Unless stated otherwise in each figure legend, the results were expressed as mean $\pm$ SD. The median was shown in heterotopic ossification datasets where non-parametric tests were applied. Each figure legend has detailed information explaining the statistical test used to compare between groups, the compared groups, the explanation of the symbols used to show significance, usually * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and the sample size of the experiment. Heterotopic ossification, evaluation, and quantification was performed blinded to mouse and group identity. Microscopy and histology representative images were selected from the total quantified images. Microscopy and histology quantified fields were randomly acquired from the plate or the region of interest. Statistical tests were performed on GraphPad Prism 9.5.

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Additional information

Author contributions

J.A.V., A.D., M.W., M.J.G., J.L.R., G.S-D. and F.V. designed the study, and analyzed and interpreted the results. J.A.V., A.D., M.W. and C.P-L. performed the experiments. J.A.V., A.D., G.S-D. and F.V. wrote the manuscript. All authors contributed to the manuscript. G.S-D. and F.V. supervised the project.

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Reviewer #1 (Public review):

Summary:

In the present study, the authors examined the possibility of using phosphatidyl-inositol kinase 3-kinase alpha (PI3Ka) inhibitors for heterotopic ossification in fibrodysplasia ossificans progressiva. Administration of BYL719, a chemical inhibitor of PI3Ka, prevented heterotopic ossification in a mouse model of FOP that expressed a mutated ACVR1 receptor. Genetic ablation of PI3Ka also suppressed heterotopic ossification in mice. BYL719 blocked osteo/chondroprogenitor specification and reduced inflammatory responses by reducing the number of fibro-adipogenic progenitors (FAPs) and promoting muscle fibre regeneration in vivo. The authors claimed that inhibition of PI3Ka is a safe and effective therapeutic strategy for heterotopic ossification.

Strengths:

Taking together previous reports on the specificity of BY718 in PI3K, it was suggested that BYL719 inhibits heterotopic ossification by reducing FAPs and promoting muscle regeneration through the PI3K pathway in vivo.

Weaknesses:

In the original manuscript, there was the possibility that BYL719 inhibited heterotopic ossification through non-specific and toxic effects rather than the PI3k pathway.

However, the authors added new data and explanations in the revision to solve the possibility. The findings of the authors would be useful and would provide an additional direction to develop a therapeutic strategy for heterotopic ossification.

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Reviewer #2 (Public review):

Summary:

Authors in this study previously reported that BYL719, an inhibitor of PI3K α , suppressed heterotopic ossification in mice model of a human genetic disease, fibrodysplasia ossificans progressive, which is caused by the activation of mutant ACVR1/R206H by Activin A. The aim of this study is to identify the mechanism of BYL719 for the inhibition of heterotopic ossification. They found that BYL719 suppressed heterotopic ossification in two ways: one is to inhibit the specification of precursor cells for chondrogenic and osteogenic differentiation and the other is to suppress the activation of inflammatory cells.

Strengths:

This study is based on authors' previous reports and the experimental procedures including the animal model are established. In addition, to confirm the role of PI3K α , authors used the conditional knock-out mice of the subunit of PI3K α . They clearly demonstrated the evidence indicating that the targets of PI3K α is not members of TGFBR by a newly established experimental method.

Weaknesses:

Overall, the presented data were closely related to those previously published by authors' group or others and there were very few new findings. The molecular mechanisms through which BYL719 inhibits HO remain unclear, even in the revised manuscript. Heterotopic ossification in mice model was not stable and inappropriate for the scientific evaluation.

The method for chondrogenic differentiation was not appropriate, and the scientific evidence of successful differentiation was lacking.

The design of gene expression profile comparison was not appropriate and failed to obtain the data for the main aim of this study.

The experiments of inflammatory cells were performed cell lines without ACVR1/R206H mutation, and therefore the obtained data were not precisely related to the inflammation in FOP.

Comments on revisions:

In the R2 version, the authors performed additional experiments using mice with inducible human R206H ACVR1A. BM-MSCs isolated from these mice were used to investigate the effect of Activin-A. The results again suggested that BYL79 inhibited the chondrogenic differentiation of BM-MSCs. However, there are still no data demonstrating the effect of BYL79 on cell growth in these in vitro experiments. In Figures 7A-D, 10 μ M BYL79 strongly inhibited the proliferation of inflammatory cells, suggesting that growth inhibition may have contributed to the results shown in Figure 5.

The main point of discussion concerns the significance of the comparisons made. The fundamental disagreement arises from the role of Activin-A in R206H cells and its effect on chondrogenic differentiation. The authors' rebuttal regarding my comments on the RNA-seq analyses should be reconsidered. The core issue lies in the interpretation of Activin-A's role in R206H cells and the distinction between chondrogenic differentiation and ossification.

A key feature of R206H mutant cells is that they respond to Activin-A by activating Smad1/5 signaling-comparable in quality to the signaling induced by BMP6 in WT cells. Another important point, as also acknowledged by the authors, is that Activin-A can transduce Smad2/3 signaling via its canonical receptor, ACVR1B. These dual signaling pathways synergistically contribute to chondrogenic differentiation in precursor cells such as FAPs. Several reports have demonstrated that the combined activation of TGF- β and BMP signaling promotes chondrogenesis more strongly than either pathway alone.

Since the PI3K α inhibition effect on HO is already known, a critical question in this study is whether BYL79 also inhibits the Smad2/3 pathway. A straightforward experiment would be to compare WT cells treated with Activin-A alone versus Activin-A plus BYL79, and to perform GO term enrichment analyses related specifically to chondrogenic differentiation, not ossification. Additionally, comparing R206H cells treated with Activin-A/BYL79 and WT cells treated with BMP6/BYL79 could help identify gene sets inhibited by BYL79 via Smad2/3 signaling. If these comparisons reveal no specific effect on genes related to chondrogenesis, the effect of BYL79 may be limited to suppression of BMP-mediated osteogenesis. Unfortunately, the authors appear to show little interest in addressing this issue.

Regarding Figure 7, the authors' rebuttal should also be reconsidered. Since the R2 version employed FOP model mice, it would have been possible to evaluate the effects of BYL79 on inflammatory cells harboring the R206H mutation. This could have enabled a more precise assessment of BYL79's influence on inflammatory signaling. While the authors repeatedly claim that BYL79's effect is not specific to any particular ligand or the presence of the FOP mutation, the role of TGF- β signaling in the development of endochondral heterotopic ossification is well recognized. Therefore, the mechanism of BYL79 should be clarified before considering its therapeutic application

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Author response:

The following is the authors' response to the previous reviews

Our revised manuscript thoroughly addresses all comments and suggestions raised by the reviewers, as detailed in our point-by-point response. To strengthen our findings, we have conducted additional *in vivo* experiments to evaluate the presence of fibro-adipogenic progenitors (FAPs) at different time points during HO formation in control and BYL719-treated mice. Our results indicate that BYL719 reduces the accumulation of FAPs and promotes muscle fiber regeneration *in vivo*. We have also expanded our discussion on BYL719's effects on mTOR signaling, further clarifying key points raised by Reviewer #1, and have addressed all minor comments.

Additionally, in response to Reviewer #2, we have employed an orthogonal and complementary approach using a new model. We conducted chondrogenic differentiation experiments with murine MSCs expressing either ACVR1^{wt} or ACVR1^{R206H}. qPCR analysis of chondrogenic gene markers (*Sox9*, *Acan*, *Col2a1*) demonstrates that Activin A enhances their expression in ACVR1^{R206H} cells, whereas BYL719 strongly suppresses their expression, regardless of ACVR1 mutational status. These new data further confirm that BYL719

effectively inhibits genes involved in ossification and osteoblast differentiation, independent of the ACVR1 mutation. We have also expanded our discussion to further clarify points raised by Reviewer #2 and have addressed all remaining minor comments.

Below, we provide a detailed point-by-point response to the reviewers' comments:

Reviewer #1:

Point 1: In this revised manuscript, the authors clearly showed that BYL719 suppressed the proliferation and differentiation of murine myoblasts, C2C12 cells, in addition to human MSCs in vitro. Furthermore, BYL719 decreased migratory activity in vitro in monocytes and macrophages without suppressing proliferation. Overall, these data suggested that BYL719 is not a specific chemical compound for cell types or signaling pathways as mentioned in the manuscript by the authors themselves. Therefore, it was still unclear how to explain the molecular mechanisms in inhibition of HO by the compound in a specific signaling pathway in a specific cell type, MSCs, contradicting many other possibilities. The authors should add logical explanations in the manuscript.

Regarding its selectivity, BYL719 is a potent and highly selective inhibitor of PI3K α . It has been demonstrated in multiple studies and in several *in vitro* kinase assay panels (Furet et al. PMID: 23726034, Fritsch et al. PMID: 24608574). The IC₅₀ or K_d values for BYL719 against PI3K α were at least 50 times lower than for most of other kinases tested. Moreover, BYL719 is also highly selective for PI3K α (IC₅₀ = 4.6 nmol/L) compared to other class I PI3K (PI3K β (IC₅₀ = 1,156 nmol/L), PI3K δ (IC₅₀ = 290 nmol/L), PI3K γ (IC₅₀ = 250 nmol/L)) (Fritsch et al). Consistent with these data, we show that, at the concentrations tested, BYL719 does not have a direct effect on any kinase receptor within the TGF- β superfamily, including ACVR1 or ACVR1^{R206H}.

Rather than blocking ACVR1 kinase activity, in our manuscript we provide evidence that BYL719 has the potential to inhibit osteochondroprogenitor specification and prevent an exacerbated inflammatory response *in vivo* (Valer et al., 2019a PMID: 31373426, and this manuscript) through different mechanisms, such as (i) increasing SMAD1/5 degradation, (ii) reducing transcriptional responsiveness to BMPs and Activin, (iii) blocking non-canonical ACVR1 responses such as the activation of AKT/mTOR. All these defined molecular mechanisms contribute to suppress HO *in vitro* and *in vivo*, as we report and explain throughout the manuscript. Selective PI3K α inhibition is at the core of the different molecular pathways described. As such, PI3K α blockade inhibits the phosphorylation of GSK3 and compromises SMAD1 protein stability, thereby altering canonical responsiveness and osteochondroprogenitor specification (Gamez et al PMID: 26896753; Valer et al PMID: 31373426). Moreover, PI3K α blockade downregulates Akt/mTOR signalling, which is critical for FOP and non-genetic (trauma induced) HO in preclinical models (Hino et al, 2017 PMID: 28758906; Hino et al. PMID: 30392977). Finally, PI3K α inhibition hampers a number of proinflammatory pathways, thereby limiting the expression of pro-inflammatory cytokines, reducing the proliferation of monocytes, macrophages and mast cells, and partially blocking the migration of monocytes. As we suggest in the discussion of the manuscript, this effect likely causes a poor recruitment of monocytes and macrophages at injury sites and throughout the *in vivo* ossification process.

Noteworthy, in our manuscript we do not refer to a “specific chemical compound for cell types”. Rather, in the Discussion we write “the administration of BYL719 prevented an exacerbated inflammatory response *in vivo*, possibly due to specific effects observed on immune cell populations.” This sentence did not intend to imply that BYL719 only affects these specific cell types, but aimed to emphasize the effects observed on those cell populations, even though systemic BYL719 may affect all populations. We rephrased it to “the administration of BYL719 prevented an exacerbated inflammatory response *in vivo*, possibly

due to the effects observed on immune cell populations.” to provide a clearer message as suggested by the reviewer. We thank the reviewer for these questions and hope that these explanations and changes in the text improve the clarity of the message.

Mesenchymal stem/stromal cells (MSCs) are osteochondroprogenitor cells that can follow distinct differentiation paths. In this study, we use these cells as an in vitro model for the study of osteochondroprogenitor specification. MSCs, and induced MSCs (iMSCs), have been widely used as in vitro cellular models of osteochondroprogenitor specification for the analysis of markers, signaling, modulation, and differentiation potential or capacity. Their use as models for this purpose has been extensively studied in wild type MSCs, and in the presence of FOP mutations (Boeuf and Richter PMID: 20959030; Schwartzl et al. PMID: 37923731).

Point 2: Related to comment #1, the effects of BYL719 on the proliferation and differentiation of fibro-adipogenic cells in skeletal muscle, which are potential progenitor cells of HO, should be important to support the claim of the authors.

We have performed additional in vivo experiments to assess the presence of fibro-adipogenic precursors (FAPs) at different time-points during HO formation in control and BYL719-treated in the mouse model of heterotopic ossification. We analyzed the number of fibro-adipogenic progenitor (FAPs) during the progression of the HO. These data are shown in the new Figure3-Figure Supplement 1. We demonstrate that BYL719 reduces the number of PDGFRA+ cells (FAPs, red) throughout the ossification process in vivo. Moreover, now we also show an enlargement of the diameter of myofibers (labelled with wheat germ agglutinin, green) when animals were treated with BYL719, indicating improved muscle regeneration and further validating the data reported as supplementary figures that were added in the first revision of this manuscript.

Point 3: BYL719 inhibited signaling through not only ACVR1-R206H and ACVR1-Q207D but also wild type ACVR1 and suppressed the chondrogenic differentiation of parental MSCs regardless of the expression of wild type or mutant ACVR1. Again, these findings suggest that BYL719 inhibits HO through a multiple and nonspecific pathway in multiple types of cells in vivo. The authors are encouraged to explain logically the use of bone marrow-derived MSCs to examine the effects of BYL719.

As detailed in main point 1, we consider that the main target, molecular mechanisms and inhibited pathways by BYL719 are specific and well characterised in other research articles and further defined in this manuscript, including the generation of PI3Ka deficient mice in an FOP background, that undoubtedly demonstrates an essential role for PI3Ka in ACVR1-driven heterotopic ossification in vivo. Altogether, we are confident that BYL719 inhibits HO through multiple and specific pathways that arise from the PI3Ka inhibition. As a systemically administrated drug, BYL719 affects the multiple types of cells in vivo that express PI3Ka. It is well known that PI3Ka is exquisitely required for chondrogenesis and osteogenesis (Zuscik et al. PMID; Gamez et al PMID: 26896753 1824619). Accordingly, throughout the manuscript we refrain from suggesting a specific effect on ACVR1-R206H cells but instead an inhibitory effect on cell number and differentiation regardless on the ACVR1 form expressed.

Similarly, as detailed in main point 1, MSCs and hiPSCs have been extensible used as in vitro cellular models of osteochondroprogenitor specification for the analysis of markers, signaling, modulation, and differentiation potential or capacity (Barruet et al., PMID: 28716551; Kan et al., PMID: 39308190).

Point 4: BYL719 clearly inhibits an mTOR pathway. Is there a possibility that BYL719 suppresses HO by inhibiting mTOR rather than PI3K? The authors are encouraged to show the unique role of PI3K in BYL719-suppressed HO formation.

As clarified above, BYL719 is a potent and selective inhibitor of PI3K α , with minimal off-target inhibition against other kinases, as it has been demonstrated in multiple studies and in several *in vitro* kinase assay panels. In the same study, while IC₅₀ of BYL719 against PI3K α was (IC₅₀ = 4.6 nmol/L), IC₅₀ against mTOR was (IC₅₀ = >9,100 nmol/L), indicating that it was not directly inhibited. mTOR is one of the well-known pathways that are activated downstream of PI3K. Therefore, there is no surprise that blocking PI3K α will block mTOR signalling. This potential effect was already demonstrated in previous publications (Valer et al., 2019a PMID: 31373426) and discussed throughout the first revision. We consider that the additive effect of mTOR inhibition and other molecular mechanisms downstream of PI3K α , including reduced SMAD1/5 protein levels, contribute to the *in vivo* HO inhibition by BYL719.

Reviewer #2:

*Point 1: It is also important to note that, in most of the data, there is no significant difference between cells with wild-type ACVR1 and those with the R206H mutation. The authors demonstrated that ACVR1 is not a target of BYL719 based on NanoBRET assay data, suggesting that BYL719's effect is not specific to FOP cells, even though they used an FOP mouse model to show *in vivo* effects.*

The main effect of R206H mutation is the gain of function in response to Activin A. For most of the responses to other ACVR1 ligands (e.g. BMP6/7), we observe a slightly increased response in the presence of the mutation (which is consistent with previous research, usually labelling RH as a “weak activating mutant” unless Activin A is added (Song et al., PMID: 20463014)). Therefore, as expected, most of the differences between WT and RH mutant cells can be observed mostly upon Activin A addition, as observed, for example, in Figure 3 of our manuscript.

We agree with the reviewer that, at the concentrations used, BYL719 does not specifically target FOP cells. However, we believe that it targets downstream pathways of PI3K α inhibition that are essential for osteochondrogenic specification, regardless of mutation status. This therapeutic strategy aligns with other experimental drugs, including Palovarotene (validated for FOP) and Garetosmab and Saracatinib (in advanced clinical trials), which target Activin A function, ACVR1 activity, or osteochondrogenic differentiation irrespective of the mutant allele. Unlike these molecules, BYL719 has been chronically administered to patients (including children) without major side effects (Gallagher et al.; PMID: 38297009), further supporting its potential for safe long-term use.

The authors should consider that the effect of Activin A on R206H cells is not identical to that of BMP6 on WT cells. If the authors aim to identify the target of BYL719 in FOP cells, they should compare R206H cells treated with Activin A/BYL719 to WT cells treated with BMP6/BYL719.

We use Activin A and BMP6, both high-affinity ACVR1 ligands, to demonstrate, as observed in figure 6, that PI3K α inhibition can inhibit the expression of genes within GO terms ossification and osteoblast differentiation. It is important to note, however, that Activin A canonical signaling receptor is ACVR1B. Since BYL719 blocks the induction of a heterotopic ossification gene expression signature common to Activin A and BMP6, in the context of the FOP mutation R206H, our results indicate that BYL719 inhibition affects a signaling pathway downstream of ACVR1, activated by either BMP6 (wild type receptor, relevant for non-genetic heterotopic ossifications) or Activin (R206H mutant receptor, relevant for FOP).

We consider that the comparison (RH ACTA BYL vs WT BMP6 BYL) would provide confounding results raised from intrinsic model differences in basal expression programs (WT vs RH), and differences in the quantitative level of signaling of the different ligands at

these specific doses. First, if we only consider SMAD1/5 signaling, Activin A and BMP6 won't have identical signaling, and differences will arise from the strength of that signaling. Secondly, in the suggested comparison we would find, mostly, all the differential gene expression promoted by Activin A canonical signaling through type I receptors ACVR1B/ALK4 in complex with ACVR2A or ACVR2B, promoting SMAD2/3 activation (in addition to the altered signaling that ACVR1-R206H could promote). Examples of differential response in pSMAD1/5 in ACVR1-WT or RH with BMP ligands and R206H with Activin A ligand, and examples of pSMAD2/3 canonical signaling in R206H cells have been described in Ramachandran et al, PMID: 34003511; Hatsell et al., PMID: 26333933).

Point 2: The interpretation of the data in the new Figure 5 is inappropriate. Based on the expression levels of SOX9, COL2A1, and ACAN, it is unclear whether the effect of BYL719 is due to the inhibition of differentiation or proliferation. The addition of Activin A showed no difference between ACVR1/WT and ACVR1/R206H cells, suggesting that these cells did not accurately replicate the FOP condition.

To gain consistency in our manuscript, we decided to use an orthogonal and complementary approach in a completely new model. We performed new experiments of chondrogenic differentiation using murine MSCs from UBC-Cre-ERT2/ACVR1^{R206H} knock-in mice. These cells, when treated with 4OH-tamoxifen, express the intracellular exons of human ACVR1^{R206H} in the murine *Acvr1* locus. Therefore, we can compare differentiation of wild type and R206H MSCs isolated from the same mice. We initiated the chondrogenic differentiation assay from confluent cells to minimize changes in cell proliferation throughout the process. These new results are shown in the new Figure 5F. Mutant (RH) cells display an enhanced chondrogenic response to activin A compared to wild type cells. The treatment with BYL719 decreased the expression of chondrogenic markers irrespective of the mutational status of ACVR1 in the cells, further supporting our previous results in this manuscript and published article (Valer et al., 2019a PMID: 31373426).

Point 3: The additional investigation of RNA-seq data provided useful information but was insufficient to fully address the purpose of this study. The authors should identify downregulated genes by comparing WT cells treated with Activin A/BYL719 and Activin A alone and then compare these identified genes with those shown in Figure 5E. Additionally, they should compare R206H cells treated with Activin A/BYL719 to WT cells treated with BMP6/BYL719. These comparisons will clarify whether there are FOP-specific BYL719-regulated genes.

We thank the reviewer for considering that RNAseq data provides useful information. As already discussed in our answer above, our results indicate that regardless of the ligand (Activin A or BMP6) and regardless of the ACVR1 mutation (WT, relevant for non-genetic heterotopic ossifications or RH, relevant for FOP), BYL719 can inhibit the expression of the genes relevant to endochondral ossification. In our opinion, this is a very relevant conclusion of this study.

We have deeply considered the strategy proposed by the reviewer, comparing “WT cells treated with Activin A/BYL719 and Activin A alone and then compare these identified genes with those shown in Figure 5E” and/or comparing “R206H cells treated with Activin A/BYL719 to WT cells treated with BMP6/BYL719”. While we have discussed why we do not consider appropriate the first comparison proposed, there are a number of reasons why we are not confident that the second comparison would provide a straightforward conclusion.

Regarding the second suggested comparison already in Main point 1, we consider that it would provide confounding results due to all the arguments detailed in Main point 1. Regarding the first suggested comparison, we also consider that it would provide confounding results. There are several reasons why we do not consider that the genes only

found in the RH comparison can be confidently considered genes that are only affected by BYL719 in RH cells.

First, the effect of BYL719 in an osteogenic-prone sample (for example, RH-ActA) is higher than the effect that we can observe in absence of this activation (for example, WT-ActA), as observed in the higher number of significantly downregulated genes in RH ActA BYL vs RH ActA comparison, compared to WT ActA BYL vs WT ActA. Similar results are observed in figure 3C, where the expressions of the genes are significantly inhibited in RH ActA compared to RH ActA BYL. This inhibition is not significantly observed in WT ActA compared to WT ActA BYL because the osteogenic expression of these genes is already very weak in the absence of ACVR1 R206H. This weak signaling of pSMAD1/5 in the absence of osteogenic signaling (RH without ligand or, especially, WT with Activin A) has already been described (Ramachandran et al. MID: 34003511). Therefore, even though the inhibition is present in both comparisons, as observed in figure 6C, the extent of the observed effect is different. Second, we are comparing a different number of DEGs for each comparison between them. If we compare the 67 downregulated genes from one comparison and 38 downregulated genes from the other comparison, the unequal list size may inflate the number of unique genes in the group with more downregulated genes. To prove these concerns, we performed the comparison that the reviewer suggested and we found, for example, that amongst the 38 differentially downregulated ossification genes in (WT_ActA_BYL vs WT_ActA) and 67 differentially downregulated ossification genes in (RH_ActA_BYL vs RH_ActA), 39 genes were only found in the RH comparison, while 10 were only found in the WT comparison, and 28 were found in both.

These effects are present, for example, when studying the ID genes, well-known downstream mediators of BMP signaling. In this case, *ID1* is downregulated in both comparisons, while *ID2*, *ID3*, and *ID4*, are downregulated only in the RH-group, despite the fact that all *ID1*, *ID2*, *ID3*, and *ID4* are similarly regulated and increase their expression with similar time curves upon BMP signaling activation (Yang et al., PMID: 23771884). Therefore, we consider that the comparisons proposed will not help us to identify specific BYL719-regulated genes relevant for FOP and/or ACVR1 R206H signaling. Again, we consider that BYL719 effect is not specific of FOP cells. Our results show that regardless of the ligand (Activin A or BMP6) and regardless of the ACVR1 mutation (WT, relevant for non-genetic heterotopic ossifications or RH, relevant for FOP), BYL719 can inhibit the expression of the genes linked to ossification and osteoblast differentiation, which could be important for the treatment of FOP and non-genetic heterotopic ossifications.

Point 4: The data in Figure 7 are not relevant to the aim of this study because the cell lines used in these experiments did not have ACVR1/R206H mutations. The authors mentioned that BMP6 is a ligand for ACVR1 and, therefore, these experiments reflect the situation of inflammatory cells in FOP. This is inappropriate and not rational. As mentioned above, the effect of Activin A on FOP cells is not identical to the effect of BMP6 in wild-type cells. The data in Figure 7 indicated that the effect of BYL719 is unrelated to the presence of BMP6, clearly demonstrating that these experiments are not related to the activation of ACVR1. In the gene expression analyses, almost all genes showed no changes with the addition of BMP6. Only TGF and CCL2 showed upregulation in THP1 cells, and the treatment with BYL719 failed to inhibit the effect of BMP6, suggesting that these experiments merely demonstrate the effect of BYL719 on inflammatory cells irrespective of the presence of the HO signal.

We consider that Figure 7 is relevant to the aim of this study. As shown in Fig. 8, treatment of FOP mice with BYL719 led to a decreased recruitment of immune cells within the FOP lesions, suggesting a direct effect of BYL719 in immune cells. This is very relevant for the FOP pathology, since flare-ups have been linked with inflammatory episodes since the very early characterization of the disease (Mejias-Rivera et al., PMID: 38672135). Given the technical

difficulties to transduce THP1, RAW264 and HMC1 cell lines with lentiviral particles carrying ACVR1 R206H, we decided to partially recapitulate ACVR1 R206H activation with recombinant BMP6 and to test the effect of BYL719 in these conditions. In these models, we found that BYL719 inhibited the expression of key genes driving immune cell activation, in a cell-type and ligand independent manner. To clarify this rationale, we have swapped Figures 7 and 8 and adjusted our conclusions accordingly. We have softened our interpretations, emphasizing the absence of the ACVR1 R206H mutant receptor in these experiments.

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